

Advances in Artificial Rabbits Optimization: A Comprehensive Review

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Abstract: This study provides an in-depth review and analysis of the Artificial Rabbit Optimization (ARO) algorithm inspired by the survival strategies of rabbits. The ARO tries to find the global solution in the search space according to the rabbits' detour foraging strategy and searches locally according to their random hiding structure. This algorithm has various advantages such as a simple structure, fast running model, easy adaptation feature, few parameters, independent mechanism in exploration and exploitation phases, transitions between phases with a specific mechanism, reasonable convergence rate, and property of escaping local optima. Therefore, it has been preferred by many researchers to solve various complex optimization problems. ARO-based studies have been published at prestigious international publishers such as Elsevier, Springer, MDPI, and IEEE since its launch in July 2022. The rates of studies in these publishers are 34%, 19%, 18%, and 15%, respectively. The remaining 14% includes papers published by other publishers. Besides, the cited studies on this algorithm are examined in four categories: improved, hybrid, variants, and adapted. Research trends demonstrate that 27%, 31%, 9%, and 33% of ARO-based studies fall into these categories.

Keywords: Artificial Rabbit Optimization, Metaheuristic Algorithm, Classification, Hybrid, Improved.

1. Introduction

Meta-heuristic (MH) algorithms are heuristic approaches designed to solve real-world problems and continue to be used more frequently and widely as time passes [1]. These algorithms are generally preferred when an objective function needs to be optimized and when exact solution methods are difficult or impossible to apply [2]. MH algorithms typically deal with problems such as screening, optimization, and finding the best possible solution. These algorithms follow an iterative mechanism to explore the solution space using the heuristic approach. Initially, an initial solution is chosen randomly, and then the solution space near this solution is exploited. The new solution obtained from the exploration is evaluated according to the optimization criterion. When a good solution is found, this solution is stored, and the discovery process continues in the next iteration. These steps are repeated simultaneously until the specified termination criterion is met or the maximum number of iterations is reached. Finally, the best solution found is considered the most optimal solution for the relevant problem [3]. These algorithms, consisting of exploration and exploitation phases, can become successful and low-cost methods with the stochastic approach. Because they constantly scan the search space in more detail and can quickly consume the resources they find [4]. The more efficiently and balanced an MH algorithm performs the tasks defined and expected in these phases, the higher its success rate and the more it can be used to solve various problems. On the other hand, according to the No Free Lunch (NFL) theory [5], due to the nature of these algorithms, they cannot always guarantee the best solution, and while they are suitable for some problems, they may not be ideal for others. Therefore, there is a need to develop and improve new algorithms. It is unclear why and when to ignore this theorem. However, this theorem is predicted

not to be ignored until someone can develop a super algorithm that is better than a random and black box algorithm for solving all real-world problems [6].

The MH algorithms try to explore solution areas efficiently, act balanced between phases and find the most appropriate solutions quickly. These algorithms are used to solve various complex problems in the real world. Examples of these application areas are feature selection [7], pattern recognition [8], image processing [9], IoT [10], mobile networks [11], vehicle routing [12], microwave [13], antenna design [14], transistors [15], image fusion [16], intelligent grids [17], energy [18], path planning [19], robotics [20], artificial intelligence [21], frame structure [22], bioinformatics [23], agriculture [24], constraint design problems [25], traffic management [26], software testing [27] and much more can be offered. Besides, the MH algorithms are divided into two categories: single-solution and population-based [28]. In single solution-based algorithms, a single candidate solution is used to improve this solution around itself, and a local search is performed. On the other hand, population-based algorithms move collectively to multiple local solutions and a single global solution in the search area, keeping possible solutions and focus by managing the process well. Therefore, while single-solution algorithms are likely to fall into the local optima trap, in the other group, the diversity of solutions and the probability of finding the global will be high since the search space is well scanned. This scenario is more valid if there is more than one local optima in a problem. Therefore, population-based algorithms generally use resources efficiently because they can be faster and find better answers, and thus, the literature mainly focuses on this type of algorithm. The MH algorithms are generally evaluated in 5 categories more accepted in the literature [29]. As shown in Figure 1, these are Evolutionary-based Algorithms (EAs), Physics-based Algorithms (PAs), Human Behavior-based Algorithms (HBAs), Swarm Intelligence-based Algorithms (SIAs), and Hybrid Algorithms (HAs). Algorithms in each category are employed by the needs of the problem and application addressed, such as single-mode and multi-mode, single-objective and multi-objective.

Darwin's theory inspires EAs. studies in this category are presented in [30-32]. PAs are inspired by the laws of physics in nature and behave randomly according to assumed physical events. Studies in this category are more challenging to develop than the other four categories because their sources of inspiration are more limited. Some examples of studies in this context are given in [33-35]. Algorithms based on many characteristics, such as human lifestyle and social behavior, constitute the third category. Some sample studies in this field are presented in [36-38]. Since each category has various features, the proposed algorithms fall into one according to their structure and nature. However, the most widely used and preferred category is SI-based algorithms. They are preferred because they are simpler, faster, and more easily inspired by nature. These algorithms generally work with the swarm mechanism. A herd means a community of dispersed individuals interacting with each other. Even though individuals work independently of each other, they are influenced by each other according to the interests of the herd, and this interactive working model has caused these algorithms to become more efficient and popular. Some works developed in this category are presented in [39-42]. In the hybrid category, multiple algorithms of the same or different categories are recommended as a single algorithm. Some examples from this category are presented in [43-47]. It is worth emphasizing that hybrid approaches are becoming increasingly widespread and are defined according to the problems.

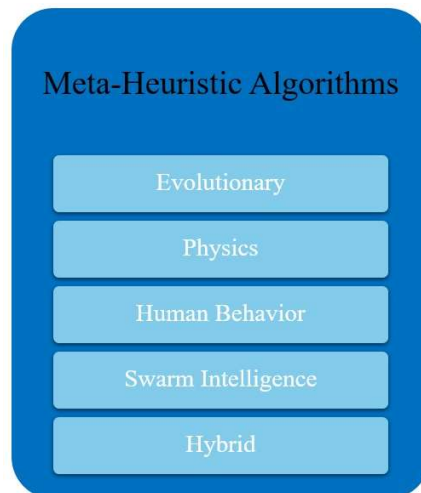


Figure 1. A classification of the MH algorithms.

1.1. Outline of the Work and Contribution

In this study, the ARO algorithm [48], one of the SI-based MH algorithms introduced by Wang et al. in 2022 and attracting much attention in the literature, is examined, and a comprehensive review study is prepared. It tries to find the optimal solution regardless of the physical nature of the problem. This algorithm is inspired by rabbits that try to survive against predators in the wild by using detour foraging, random hiding, and energy shrink strategies. The ARO has a simple structure, fast-running model, easy adaptation feature, few parameters, independent mechanism in exploration and exploitation phases, reasonable convergence rate, and property of escaping local optima. Therefore, it is an algorithm that is low-cost, flexible, easy to implement, and does not require a derivation mechanism. For these reasons, many researchers have started using this algorithm to solve various complex problems. In the first study introduced, this algorithm was compared with Particle Swarm Optimization (PSO) [49], Atom Search Optimization (ASO) [33], Differential Evolution (DE) [50], Gravitational Search Algorithm (GSA) [51], Cuckoo Search (CS) [52], Artificial Bee Colony (ABC) [53], and Teaching Learning Based Optimization (TLBO) [36] algorithms on 31 benchmark functions and six complex engineering problems. According to the analysis, it has been shown that the ARO algorithm is quite competitive and effective in overcoming complex and nonlinear problems with unknown and limited search space. Therefore, this paper focuses on the ARO algorithm, which has quickly gained attention in the literature due to the advantages described, and all studies based on this algorithm have been thoroughly reviewed so far. Thus, this review paper provides a comprehensive overview of ARO, its variants, and applications. The main contributions of this review are:

- A review of the statistics of ARO-based studies published by different publishers.
- A review of the variants and classification of the ARO algorithm in four groups (Improved, Hybrid, Variants, and Adapted).
- A review of the role and achievements of the ARO algorithm in various applications and problems.

The rest of this paper is organized as follows. Section 2 briefly explains the working mechanism of the ARO algorithm. Section 3 presents the statistics of ARO-based studies from different perspectives. Sections 4 and 5 discuss the classifications and application types of the ARO algorithm with the advantages and disadvantages of existing studies. Section 6 presents an analysis of ARO's performance. Section 7 covers general discussions. The conclusion and future works will take place in the last section.

2. ARO: Artificial Rabbit Optimization

Artificial Rabbit Optimization (ARO) [48] is a new swarm intelligence-based optimization algorithm inspired by rabbits' hiding and foraging behavior in nature. In this regard, the algorithm uses hiding, foraging, and energy shrink strategies. This algorithm, which includes foraging and hiding features, simulates the exploration and exploitation phases. Additionally, the energy reduction strategy represents transitions between stages. In this section of the paper, the methods of the ARO algorithm are summarized with mathematical models.

2.1. Initialization

Like other SI-based algorithms, the herd population must first be created in ARO. The initial position of this randomly generated population is represented in Equation 1. Each individual (rabbit) in the population tries to solve a relevant problem using a defined fitness function. The agent that finds the best solution in each iteration directs the others, and in this way, the algorithm improves the answers they find as the iterations progress. The use of each agent in the corresponding function is described in Equation 2. The ARO algorithm is represented by a two-dimensional matrix consisting of agents and problem dimensions (Figure 2).

$$X = [X_1, X_2, \dots, X_i \dots X_n]^T = \begin{bmatrix} X_{11} & \dots & X_{1d} \\ \vdots & X_{ij} & \vdots \\ X_{n1} & \dots & X_{nd} \end{bmatrix} \quad (1)$$

$$F_X = \begin{bmatrix} f(X_{11}) & \cdots & X_{1d} \\ \vdots & & \vdots \\ f(X_{n1}) & \cdots & X_{nd} \end{bmatrix}_{n \times d} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix}_{n \times 1} \quad (2)$$

Where ‘ n ’ represents the number of rabbits and ‘ d ’ indicates the dimensions of the problem. Also, ‘ X_{ij} ’ is the position of agent ‘ i ’ in dimension ‘ j ’. The $[X_{i1}, X_{i2}, \dots, X_{id}]$ represents the positions of the i^{th} agent.

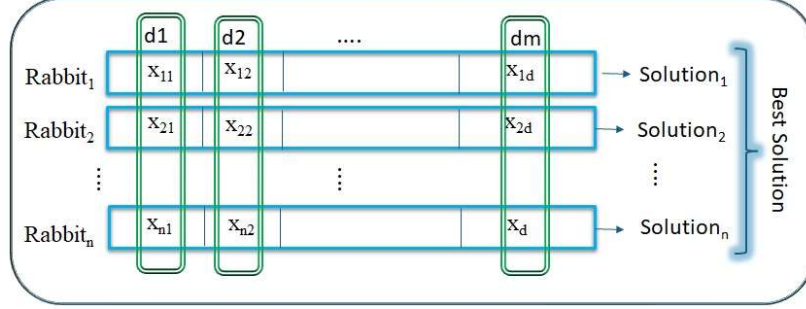


Figure 2. Population of ARO algorithm.

2.2. Detour foraging mechanism (Exploration)

In their devious foraging strategy, Rabbits eat grass near other rabbits' burrows, which may leave their burrow undiscovered by predators. In general, rabbits' nature is that they scan the distance and ignore those nearby when they hunt. Therefore, they randomly consume herbs elsewhere rather than within their territory, and accordingly, this approach is called the detour foraging strategy. Rabbits are more likely to disturb the area close to the source while searching for food. Based on this property, the algorithm prefers to update the position of each search individual relative to another search individual randomly selected from the swarm and contribute to a perturbation. The mathematical model based on this strategy is presented in Equation 3. According to this equation, individuals allow other rabbits to randomly search for food based on location and change their homes accordingly. In this way, rabbits are expected to be successful in global search by visiting others' nests instead of their own. This phase can help ARO avoid local optima and perform global search in the search space.

$$\vec{V}_i(t+1) = \vec{X}_j(t) + R \cdot (\vec{X}_i(t) - \vec{X}_j(t)) + \text{round}(0.5 + (0.05 + r_1) \cdot n_1); i, j = 1 \dots n, i \neq j, n_1 \sim N(0,1) \quad (3)$$

$$R = L \cdot C \quad (4)$$

$$L = \left(e - e^{\left(\frac{t-1}{T}\right)^2} \right) \cdot \sin(2\pi r_2) \quad (5)$$

$$C(k) = \begin{cases} 1 & \text{if } k = g(l) \\ 0 & \text{otherwise} \end{cases}; k = 1 \dots d, l = 1 \dots [r_3 \cdot d], g = \text{randperm}(d) \quad (6)$$

Where, ‘ $V_i(t+1)$ ’ denotes the candidate position of ‘ i^{th} ’ rabbit at ‘ $t+1$ ’ time (next iteration), ‘ X_i ’ and ‘ X_j ’ indicate the position of ‘ i^{th} ’ and ‘ j^{th} ’ rabbits. The ‘ t ’ and ‘ T ’ represent the current iteration and maximum iteration. ‘ r_1 ’, ‘ r_2 ’, and ‘ r_3 ’ are random numbers between $[0, 1]$. The ‘ round ’ specifies rounding to the nearest integer, while ‘ randperm ’ functions to return a random permutation of the integers 1 through ‘ d ’. Also, ‘ n_1 ’ is subject to the standard normal distribution. ‘ R ’ is a parameter that simulates the running characteristic of rabbits (Equation 4). ‘ L ’ and ‘ C ’ parameters play a role in this calculation. ‘ L ’ represents the movement speed (running length) in the position update (Equation 5). As the iterations progress, this parameter will have smaller values so individuals can move with shorter steps. This behavior is visually illustrated in Figure 3. ‘ C ’ indicates the mutation that individuals undergo in foraging. In this way, it was assumed that some individuals would randomly select regions and thus contribute to the global search (Equation 6).

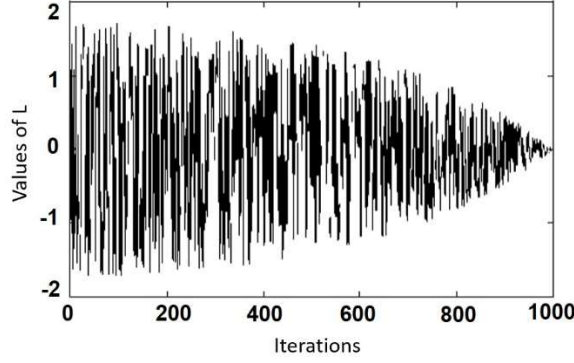


Figure 3. Dynamic behavior of L during 1000 iterations [48]

2.3. Random hiding (Exploitation)

The ARO algorithm creates random nests for individuals in all directions in each iteration, inspired by the ability of rabbits to develop continuous and random nests. In this way, rabbits enter the underground by randomly choosing one of the 'd' burrows to hide in and make other passages through tunnels. This allows them to escape from predators. As a simple example, the model in which the " i^{th} " rabbit is moved to the " j^{th} " nest is presented in Equation 7. Accordingly, 'd' burrows are created along each dimension in a rabbit's neighborhood.

$$\vec{b}_{i,u}(t) = \vec{X}_i(t) + H \cdot g_u \cdot \vec{X}_i(t); i = 1 \dots n, j = 1 \dots d \quad (7)$$

$$H = \frac{T-t+1}{T} \cdot n_2; n_2 \sim N(0,1) \quad (8)$$

$$g_u(k) = \begin{cases} 1 & \text{if } k = [r_5 \cdot d] \\ 0 & \text{otherwise} \end{cases}; k = 1 \dots d \quad (9)$$

Where, ' $b_{i,j}$ ' represents a randomly selected hiding burrow. In calculating this, the variables ' H ' and ' g_u ' are obtained according to Equations 8 and 9. ' r_4 ' and ' r_5 ' are random numbers between [0, 1]. Moreover, ' H ' is the hiding parameter, which is reduced linearly from 1 to $1/T$ by a random perturbation across iterations [54] and is calculated by Equation 8. According to this parameter, while the number of nests is high in the first iterations, the number of neighborhoods decreases inversely as the iterations progress. Therefore, the position updating of the rabbits in this phase is comprehensively presented in Equation 10. Based on this, the ' i^{th} ' individual updates its position by randomly selecting one of the 'd' burrows.

$$\vec{V}_i(t+1) = \vec{X}_i(t) + R \cdot (r_4 \cdot \vec{b}_{i,u}(t) - \vec{X}_i(t)); i = 1 \dots n \quad (10)$$

As a result, after one of the devious foraging or random hiding methods has been successfully carried out, the position update of the ' i^{th} ' rabbit is done according to Equation 11. Accordingly, if the fitness value of a rabbit's candidate position is better than its current location, the next position of that rabbit will be the candidate position. Equation 3 or 10 will be applied, depending on the situation. Otherwise, the rabbit maintains its position.

$$\vec{X}_i(t+1) = \begin{cases} \vec{X}_i(t) & \text{if } f(\vec{X}_i(t)) \leq f(\vec{V}_i(t+1)) \\ \vec{V}_i(t+1) & \text{otherwise} \end{cases} \quad (11)$$

2.4. Transition management and complexity analysis

In the ARO algorithm, searching for food predominates in the first iteration, while random hiding comes to the fore in the last iteration. In this case, parameters such as ' L ' and ' C ' play an essential role. However, the most critical and practical parameter depends on the parameter ' A ' presented in Equation 12. In this algorithm, it is called energy shrink. This fact is due to the rabbit's decreasing energy over time in nature. Therefore, in the algorithm, an energy factor is defined by imitating this technique, and accordingly, transitions between the exploration and exploitation phases are achieved.

$$A(t) = 4 \left(1 - \frac{t}{T}\right) \ln \frac{1}{r_6} \quad (12)$$

Where ' r_6 ' is a random number in $[0, 1]$. According to this equation, the energy factor decreases as the iterations increase. In this case, if $A(t) > 1$, rabbits will be forced to explore; otherwise, they will try to exploit (Figure 4).

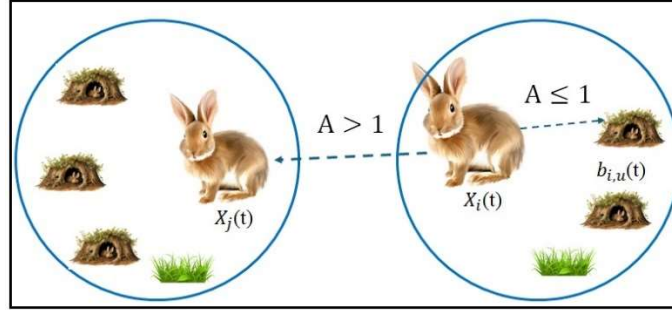


Figure 4. Exploration and exploitation behavior of ARO based on energy shrink

2.5. Pseudocode, flowchart, and complexity analysis of ARO algorithm

This section presents the ARO algorithm's working mechanism as a flow diagram and pseudocode in Figure 5 and Algorithm 1, respectively.

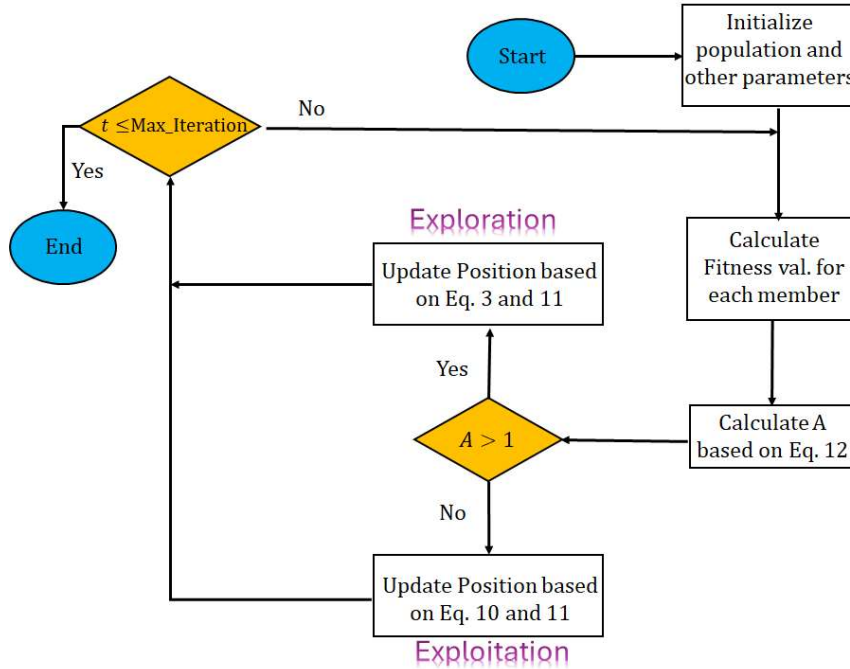


Figure 5. The flowchart of the ARO algorithm.

Algorithm 1. The pseudocode of the ARO algorithm

```

Initialize a set of rabbits  $X_i$  (solutions)
Calculate fitness values and select  $X_{best}$ 
While  $t < T$ 
  For each individual
    Calculate the energy factor  $A$  using Equation 12
    If  $A > 1$ 
      Choose a rabbit randomly from other individuals.
      Calculate  $R$  using Equation 4
      Perform detour foraging using Equation 3
      Calculate the fitness
      Update the position of the current individual using Equation 11
    Else
      Generate  $d$  burrows and randomly pick one as hiding using Equation 10
      Perform random hiding using Equation 7
      Calculate the fitness
      Update the position of the individual using Equation 11
  End If
End While

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```

Update the best solution found so far  $X_{best}$ 
End For
End While
Return  $X_{best}$ 

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On the other hand, the complexity calculation of ARO relates to the maximum number of iterations (T), the number of search agents (n), and the variable size of the problems (d). It is equivalent to $O(Tnd+Tn+n)$. This means $O(N^2)$ in classical notation.

3. Statistics of ARO-based Studies

This section reviews statistics about relevant studies based on the ARO algorithm. Various research has been conducted to solve optimization problems with ARO since July 2022. The statistics of all studies offered by ARO are examined in this paper to demonstrate this more clearly. Based on available citation context data and extracts from 248 cited items, ARO was either used for background/discussion/comparison or directly utilized in citing articles. In this regard, a grouping has been prepared according to the percentage of papers in various publishers and the annual number of publications. These publications are taken into account until May 2024. Detailed information about the publishing houses and the total citations made during this period is shown in Figure 6. Of these 248 citations, 18 are published in 2022, 130 studies are published in 2023, and 100 consist of publications until May 2024. These studies include journal and conference papers, book chapters, and graduate theses. The analysis comprises over five groups, considering the publishers where the cited articles were published. Eighty-four of these publications were published in Elsevier, 47 in Springer, 44 in MDPI, 38 in IEEE, and 35 in other publishers. According to the review, 34% of manuscripts have been published in Elsevier, 19% in Springer, 18% in MDPI, 15% in IEEE, and 14% in other publishers. These analyses show that the highest percentage of published papers belong to Elsevier. Moreover, approximately 45% of these citations used the algorithm directly to solve a specific problem or modified the original algorithm. The analysis demonstrates that the trend of the ARO algorithm is increasingly accelerating. In fact, through May 2024, based on publicly available data, approximately 40% of all historical citations are from these five months. The ARO algorithm tends to be used primarily on papers published in prestigious peer-reviewed journals.

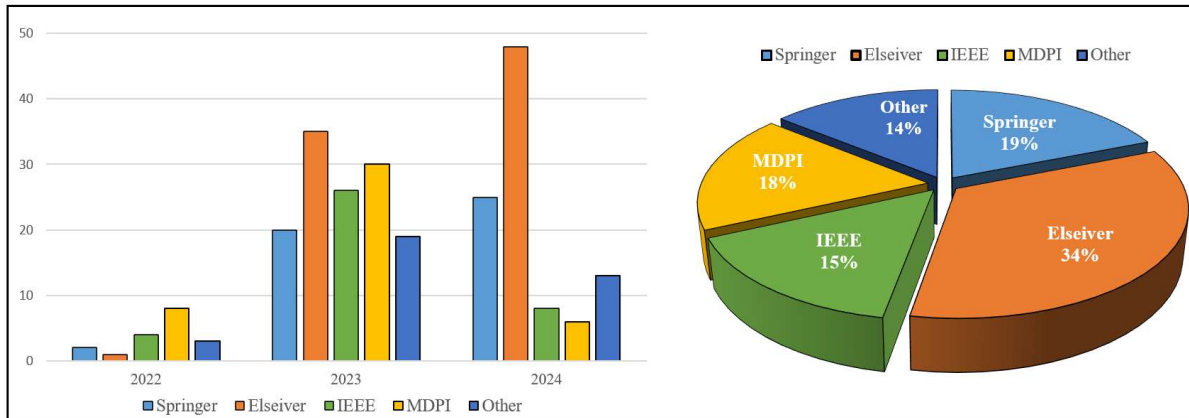


Figure 6. The number and percentage of ARO-based papers published in various publishers.

Figure 7 presents the index information for the cited references. The groupings prepared are based on the Web of Science (JCR) quadrants. Apart from this, indices such as ESCI and Scopus were evaluated in another group called other indexes. This research shows that 87% of these citations consist of SCI-level publications. 38% of the citations were publications in journals in Q1. Also, 27% were in Q2, 13% were in Q3, and 9% were in Q4. The remaining 13% consists of publications in other indexes. Besides, the contribution and importance of ARO in the literature is demonstrated by the fact that 95 studies of all citations are in Q1, which is growing. Another issue that draws attention here is the significant increase in citations in the Q1 quarter. This shows that the relevant algorithm is used by researchers in higher-quality studies and journals. In addition, the analysis results of cited papers by countries and regions are presented in Figure 8. A detailed examination of the variations and applications of this algorithm, which attracts the attention of researchers, is carried out in the following sections.

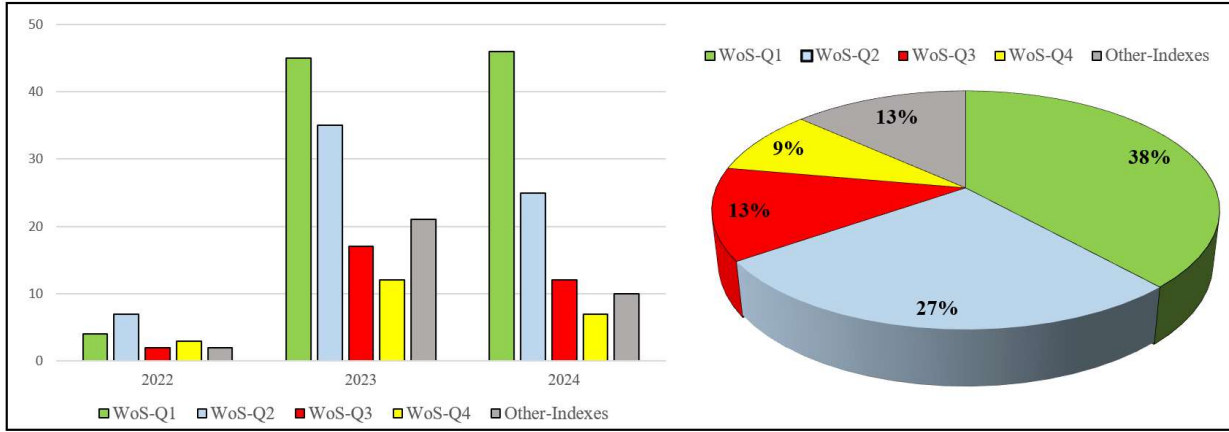


Figure 7. Index information of publications referring to the ARO algorithm.

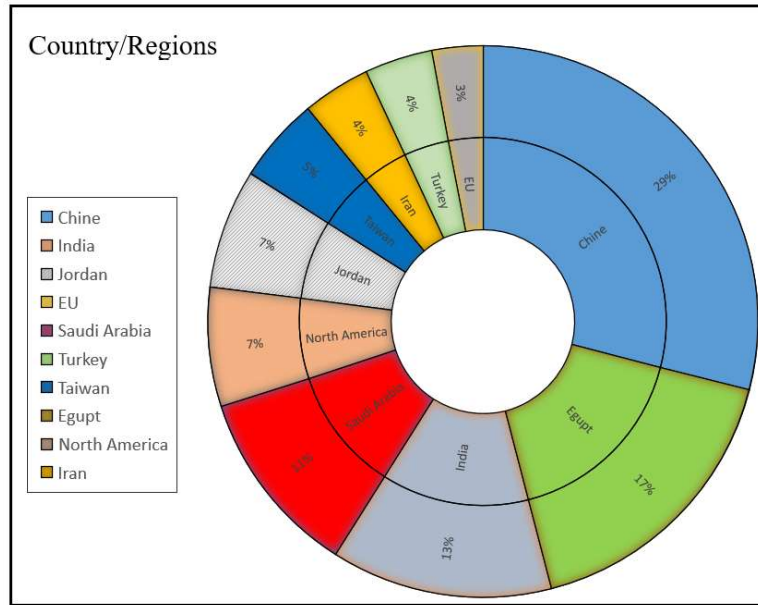


Figure 8. The percentage of total citations is based on various countries/regions.

4. Classification and Applications of the ARO

This section examines studies based on the ARO algorithm in literature. Among the referenced studies, those that use the ARO technique and its features for different purposes or suggested improvements on this algorithm are discussed. In this regard, a classification covering all studies has been defined. Accordingly, all studies are examined in four different categories. **Adapted**, **Variants**, **Hybrid**, and **Improved** classes (Figure 9). According to the research, 101 studies out of 248 citations fit into these classes. Other citations reference ARO for background, discussion, or comparison. In the *Adapted* class, the standard (original) ARO algorithm is employed to solve a specific problem. In the studies in this category, no changes to the algorithm are defined; only the appropriate fitness function is defined to run the algorithm. The *Variants* category has been examined in two subclasses: binary model and multi-objective optimization version. Moreover, studies marked as hybrid mean a new method combining ARO and another technique or algorithm. In this regard, methods combined with ARO have been categorized from three perspectives (metaheuristic, deep learning, and machine learning). The last category is the modifications and improvements made to the ARO algorithm with various strategies. The main reason for these studies is either to eliminate some of the deficiencies of ARO or to make modifications to solve specific problems more optimally. As a result of detailed examination, studies in this category used 25 different strategies. The percentage of studies in these categories is presented in Figure 10. All these 101 publications used ARO in other ways to solve various optimization problems. The problems and applications addressed by existing studies are explained in Section 5.

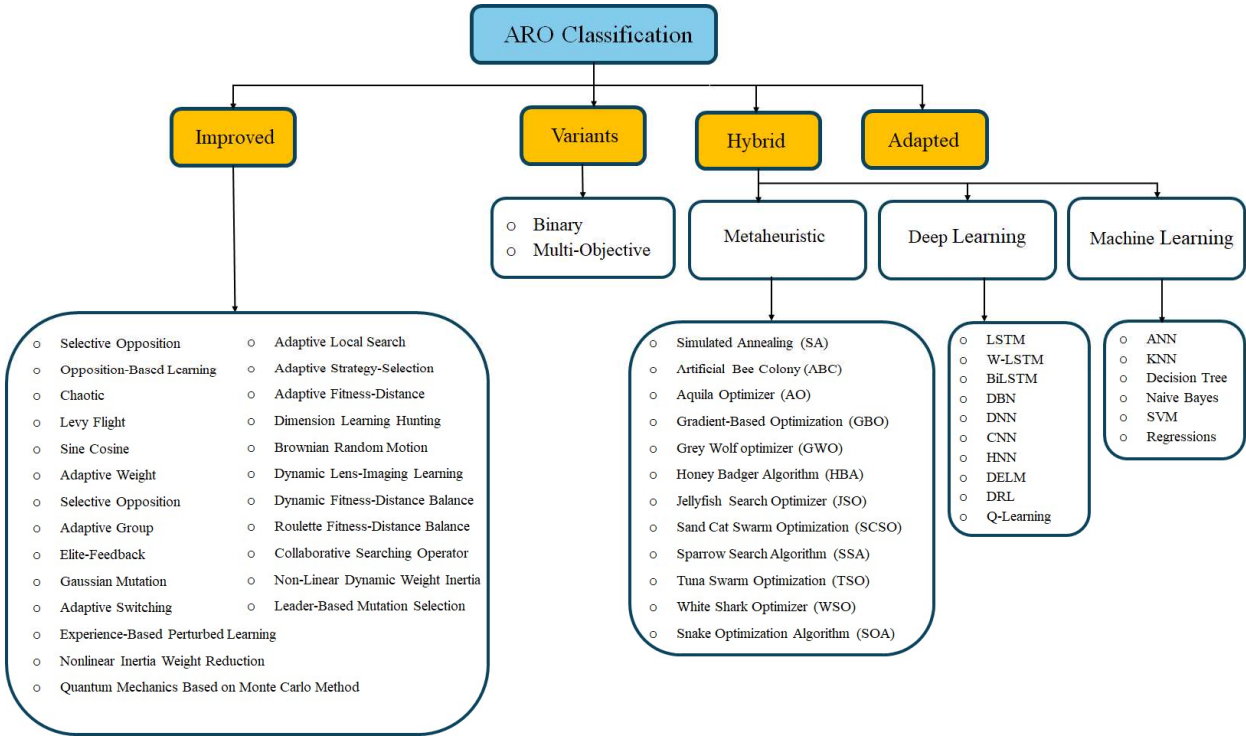


Figure 9. ARO algorithm's classification.

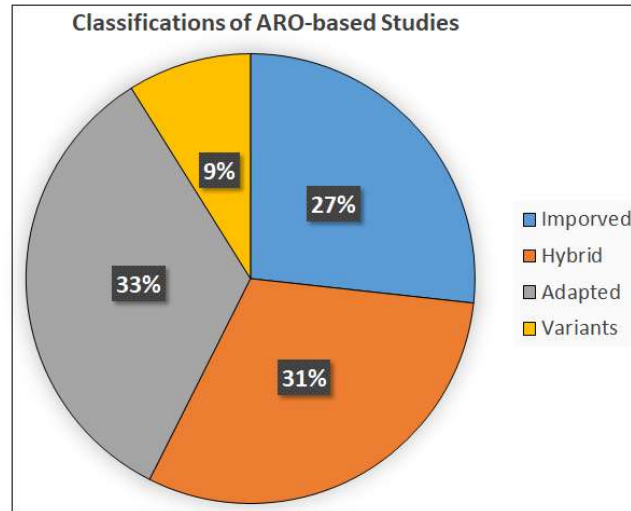


Figure 10. Percentage distribution of classification of ARO-based studies.

4.1. Hybrid class

This section examines publications focusing on hybridizing the ARO algorithm with various methods. Relevant studies are evaluated under three subheadings: MH, Machine Learning (ML), and Deep Learning (DL).

4.1.1. Metaheuristic-based studies

According to the classification, in the meta-heuristic category, the ARO algorithm is combined with Simulated Annealing (SA), Artificial Bee Colony (ABC) [53], Aquila Optimizer (AO) [55], Gradient-Based Optimization (GBO) [56], Grey Wolf Optimizer (GWO) [2], and White Shark Optimizer (WSO) [59] algorithms.

In [60], a hybrid algorithm based on the ABC and ARO is proposed in the IEEE-26 bus system to solve the economic distribution problem. The primary purpose of this algorithm is to find the optimum power at the least cost. In this study, the ABC is used for the exploration phase, and the ARO is used for exploitation tasks. It has been claimed that the proposed hybrid algorithm successfully reduces the cost of electricity generation. In the related study, since the

ARO algorithm has preferable exploitation mechanism while approaching the optimum value and, on the other hand, the ABC algorithm has a robust exploration strategy, the strengths of these two algorithms were employed together to solve the related problem. In the related study, since ARO is relatively new, it is envisaged to be tested and compared with a different approach regarding efficiency and robustness. According to the analyses, the proposed ABRO hybrid algorithm has faster convergence with a better minimum value and a lower std. In this study, in addition to the economic dispatch problem, analyses were also made on the four benchmark test functions, and the performance of the proposed hybrid algorithm was compared with ABC and ARO. According to the results, the ABRO algorithm showed superior performance, and ARO came in second.

Another study [61] focused on global optimization benchmark problems. In this study, the AO is employed in the exploration phase, and the ARO is applied in the exploitation phase. The AO has been included to make ARO's reconnaissance ability more satisfactory. This algorithm has demonstrated exemplary accuracy, convergence speed, and robustness and has been applied to industrial engineering design problems and the Photo-Voltaic (PV) model parameter determination problem. This study emphasizes that ARO has very competitive business potential, but its search capability should be more satisfactory. In this context, a new hybrid algorithm is proposed using the advantages of both algorithms to provide an approach with high search capability. For performance measurement, 23 classical benchmark functions and the IEEE CEC2019 test suite were used, and the results were compared with a total of 9 algorithms, such as AO, ARO, GWO, and WOA. The analyses were made under seven headings: convergence behavior, complexity, and scalability. In light of the information obtained and presented, the ARO-based hybrid algorithm exhibited the best performance, while ARO came in second. According to the experimental results, the proposed method (CHAOARO) can reliably solve various problems with high accuracy rate and convergence speed.

In [62], a hybrid approach has been proposed to find the optimal value for IEEE 30-Bus, IEEE 57-Bus, and IEEE 118-Bus test systems on the Optimal Reactive Power Dispatch (ORPD) problem in electric networks. This study combines the advantages of ARO and the GBO algorithm. The GBO has been used to better explore the search space with a set of vectors and two elementary operators. These operators are the Gradient Search Rule (GSR) and local Escape operators (LEO). According to the results obtained from the simulations, the accuracy and stability of the proposed hybrid algorithm in solving the relevant problem have been confirmed. In this study, a hybrid algorithm is presented using the GBO algorithm to improve the discovery capability of the ARO algorithm and to accelerate the convergence rate. Tests are performed on benchmark functions and several engineering problems (different IEEE Bus test systems), and the results are compared with ARO, GBO, Pelican Optimization Algorithm (POA), and Northern Goshawk Optimization (NGO) algorithms. The analyses are made under the titles of optimum value finding scale, convergence analysis, Wilcoxon's rank, and Friedman tests. According to these additional tests, the classical ARO algorithm is ranked fourth among the comparisons. However, classical ARO was ranked second according to the results on different IEEE Bus test systems. Again, the proposed hybrid approach (AROGBO) exhibited the best performance in this category of problems.

In another study in this category, a hybrid algorithm based on GWO and ARO was designed to detect structural damage in bridge structures [63]. Their authors proposed a new switching mechanism using the hunting mechanism in GWO to make the transitions between exploration or exploitation of ARO more efficient. In this way, the probability of falling into a local minimum will be reduced, and as a result, less accurate optimization results will be obtained. Additionally, it aims to reduce the computational time investment ARO brings. Two different scenarios were considered for computational time analysis. According to the presented results, the hybrid approach (IARO) improved around 20% in one scenario and 4% in the other compared to the standard ARO. The proposed algorithm tries to find the optimal value for beam structure and steel truss bridge problems. As a result of the study, improvements were observed in determining the exact location of the damage, its severity, and the calculation process. This study claims that the IARO algorithm minimizes computational costs to a certain extent while maintaining a competitive convergence speed and providing good performance.

In [64], a hybrid algorithm based on the ARO and WSO was suggested for the model of two PV cells and six PV modules. This study aims to obtain more effective performance by working independently of WSO and ARO algorithms and using their results together. In this regard, the relevant algorithm for parameter extraction of PV cells and modules is proposed. This study uses 7 MH algorithms such as ARO, POA, and TSO for comparison. According

to various analyses, the proposed hybrid algorithm exhibits the best performance while the classical ARO algorithm follows it. A brief of the studies conducted in this category is presented in Table 1.

Table 1. Hybrid MH studies with ARO.

Study	Focus on	Role of ARO	Hybrid with
[60]	Economic Dispatch Problem	To find the optimum power with the least cost	ABC for improving exploration
[61]	Global optimization	To find optimum values over benchmark functions and classical engineering problems	AO for improving exploration.
[62]	Optimal reactive power dispatch	To find the optimal value for IEEE 30, 57 and 118-Bus test systems	GBO for improving exploration
[63]	Identifying structural damage in bridge structures	To find the optimal value for steel truss bridge and beam structure problems	GWO for better balancing switching phases
[64]	PV cells	For modeling of two PV cells and six PV modules	WSO for getting more effective performance

4.1.2. Machine Learning-based studies

In this section, studies that combine the ARO algorithm and ML methods are examined. In [65], a design has been made for environmentally sustainable Rubberized Concrete (RC). This study has hybridized the ARO, Honey Badger Algorithm (HBA) [57], and Tuna Swarm Optimization (TSO) [58]. This study took advantage of the flexibility of the TSO algorithm in adaptation to various problem scenarios and the dynamic foraging behavior of HBA. It also desired to benefit from ARO's balanced behavior mechanism. The related study presented multiple analyses to evaluate the proposed approach and model. This algorithm tries to find the optimum compressive strength and maximize the rubber and cementitious substitute material content. The results show that it is possible to create RC with lower environmental impact and potential cost savings by determining the optimum combination of rubber, cement, and other additives. In [66], an Artificial Neural Network (ANN)-based model was developed to make efficient use plans by focusing on efficient energy consumption in buildings. In the pre-training of this model, the ARO algorithm was used to optimally solve the parameter initialization problem. This algorithm is also employed to optimize the parameters of ANN. According to the analysis, the ARO-ANN model provided significant advantages with a higher accuracy in solving the problem of energy supply and demand compared to other models. In the study, analyses were conducted on five different models, and according to the results, the proposed framework achieved an accuracy increase of approximately 6.11% with a R^2 of 98.87%. In [67], the primary focus is on traffic flow data prediction. This topic is valuable for traffic managers and participants who want to control traffic congestion and plan trips. In this context, two different ANN-based models have been developed. This study has applied the ARO to find optimal values for Kernel Extreme Learning Machine (KELM) parameters. The proposed ARO-KELM is used to predict all decomposed components. On the other hand, Variational Mode Decomposition (VMD) developed with the Northern Goshawk Optimization (NGO) algorithm has been tried to be improved. Various results emerged in this study, where analyses were made on two different scenarios and data sets. Accordingly, the ARO-based method exhibited the best performance with the smallest margin of error. The authors claimed that the proposed model is superior to other comparison models at a 99% confidence level. The [68] proposes an ANN-based model that predicts power usage in DC-AC hybrid supply systems. This issue is critical in microgrid energy management systems. Using profile data, the corresponding model has been used to indicate the charge and discharge of energy storage devices across multiple distribution network power states. The ARO algorithm was used to reduce the error rate in the predictions made by the model, and in this regard, this algorithm was used to fine-tune the ANN. According to the results, this ARO-ANN model has been claimed to be the most efficient in the literature, with an accuracy rate of 99.55%.

Another study in this category focused on the damage detection of lattice structures [69]. The proposed method uses parallel vectorized data and combines the corresponding model with an ANN to improve the accuracy and efficiency of the damage detection process. The ARO algorithm was used to optimize the weight value of the ANN to increase the model's performance. According to the study results, the proposed method has better accuracy, R, and MSE values than the traditional ANN. Also, regarding computational cost, it is claimed to run in 23% less time compared to the original ARO. In [70] is aimed to maintain a stable DC bus voltage and ensure balanced production and consumption for managing energy in the DC microgrid. In this regard, an ANN-based model has been developed, and the ARO

algorithm has been utilized to optimize the weight value of the model. This study combines the isolated DC microgrid regulator with two approaches. Incremental conductance Maximum Power Point Tracking (MPPT) technique has been used to maximize power drawn from PV sources. On the other hand, ARO-ANN control is proposed to manage the power modulator in the storage scheme. In this study, the comparison between the proposed method and the conventional method was made on the battery and supercapacitor power parameters, and according to the results, the proposed method demonstrated superior performance, albeit by a small margin. In [71], it is focused on controlling and managing the intense patient flow in Emergency Departments (ECs). Additionally, preventing patients with viral diseases from being transported from the EC to other areas is the focus of this study. The ELM model is trained to track the target audience and separate the data into many classes. The ARO algorithm is employed for the optimal selection of feature subsets. This study used actual data from KSA hospitals in Saudi Arabia. According to the simulation results in this study, the proposed approach achieved maximum accuracy of 93.22% and 94.05% on the used Cleveland and Statlog datasets, respectively, and offered better performance than other methods. The [72] study focuses on botnet attacks in IoT systems. Therefore, the authors introduce a wrapper feature selection technique using the ARO and Decision Tree (DT) algorithm to detect various types of botnet attacks. In this study, N-BaIoT datasets were employed as a testbed for the design of this model. In this study, the ARO algorithm was applied to select only high-priority features, and the output from this stage was used on the DT side to model design. The accuracy of this proposed model is claimed to be 99.89%, and it has demonstrated successful behavior in effectively reducing the feature area. In [73], an ANN-based model has been developed for suitable and efficient antenna design for devices in environments with many complex electromagnetic waves. The ARO algorithm was used to find the optimum weights of this model. The relevant model was compared with two MH-based models, and the results showed that the proposed ARO-ANN model had a higher prediction accuracy.

In [74], a classifier model based on ANN was developed to achieve higher performance in analyzing emotions in implicit aspects found in high-quality training datasets. This study's primary focus is the Robust Random Vector Functional Link Network (RVFLN). The ARO is applied to optimize this model's input weights and hidden layer biases. The proposed model was tested on three datasets, and it was claimed to be better than other methods in terms of F1 score, sensitivity, classification accuracy, error rate estimation, and kappa statistics. The [75] focused on designing Concrete-Filled Double-Skin Steel Tubular (CFDST) columns, an essential issue in civil engineering. It is emphasized that factors such as geometry, material properties, and loading conditions be considered for optimized design. In this regard, an innovative approach strengthens the synergy between Gradient Boosting Regression (GBR) and three MH optimization algorithms. The ARO, one of the MH-based algorithms used, was applied to fine-tune the hyperparameters of the prediction model, and according to the results, its performance was better than that of the other two MHs used. In [76], the aim was to estimate the Optimum Moisture Content (OMC) of soil stabilizer mixtures and to develop a comprehensive and precise model by considering various related parameters. This model is based on Support Vector Regression (SVR). The ARO is used to find optimal hyperparameters of the SVR model and validate the model by analyzing OMC samples from various soil types obtained from previous stabilization test results. The results prove that the proposed hybrid model has good sensitivity and reliability and performs about 10% better than other existing methods. This method offers a promising technique in various related engineering applications. A summary of ML-based studies is presented in Table 2.

Table 2. Hybrid ML studies with ARO.

Study	Focus on	Role of ARO	Hybrid with
[65]	Concrete design	To find optimal values of RC design	ANN and CatBoost
[66]	Energy prediction for buildings	To optimize ANN	ANN, MLP
[67]	Traffic flow data prediction	To find optimal values for KELM parameters	NN and KELM
[68]	Microgrid energy management system	To fine-tune ANN	ANN
[69]	Damage detection of truss structures	To optimize the weight value of the ANN	ANN
[70]	Isolated DC Microgrid System	To optimize the weight value of the ANN	ANN
[71]	Emergency department monitoring and medical data classification	To the optimal selection of feature subsets	ELM
[72]	IoT Botnet attacks	To select only the high-priority wrapper features	DT

[73]	Antenna modeling	To find the optimal weights of ANN	ANN
[74]	Sentiment analysis	To tune hyperparameters for accurate classifying by MLP	RVFLN
[75]	Concrete-filled double-skin steel tubular	To tune hyperparameters of the predictor (GBR) model	GBR
[76]	Optimum moisture content of soil-stabilizer mixtures	To find optimal hyperparameters of the SVR model	SVR

In addition to ML and MH-based hybrid studies of the ARO algorithm, in some studies in the literature, this algorithm has been combined with an MH algorithm and applied in an ML-based method. In this category, MH algorithms used as hybrids are GBO, HBA, Jellyfish Search Optimizer (JSO) [77], Sand Cat Swarm Optimization (SCSO) [78], and Salp Swarm Algorithm (SSA) [79]. These studies based on both MH and ML were examined separately. In [80], a Naïve Bayes (NB)-based model has been developed to accurately predict the maximum dry density of soil stabilization mixtures. In this regard, a hybrid algorithm based on GBO-ARO is proposed to tune the parameters of this model. According to the results, with this hybrid approach, the accuracy of the models has increased and can provide more reliable predictions. This outcome is essential to create a solid and long-lasting infrastructure in the construction industry. According to the results, the R^2 value of the proposed model is 0.9903, and the RMSE value is 34.563. In [81], a reliable prediction model has been created to accurately determine Compressive Strength (CS) and Flexural Strength (FS) parameters when designing reinforced concrete structures. In this regard, the CATBoost Regression (CAT) model was used to predict the mechanical properties of concrete materials. ARO and HBA algorithms were combined to optimize and fine-tune this model. It aims to obtain more effective results with a more diverse population by using the advantages of these two algorithms. According to the results, this hybrid model works with fewer errors and higher accuracy. In [82], an NB-based model is proposed to increase students' academic performance, and ARO and JSO algorithms are combined to tune the parameters of the developed model. The proposed method was compared with the NBS model and provided better performance with a rate of 1.03% in the F1 score. In [83], it has been emphasized that audits should be classified according to various types or categories to manage and report audit activities effectively. A Support Vector Machine (SVM) and ANN-based model are proposed in this audit data to eliminate noise and inconsistencies in the processing stages. The ARO and SCSO algorithms were used for feature extraction in the SVM and ANN. In comparing the hybrid models consisting of these four scenarios, it has been shown that the ARO-based SVM model is better than the others. In [84], an ANN-based temperature compensation model is proposed to improve the detection accuracy of laser methane sensors in wide-temperature application environments. In this study, the ARO was applied to improve the SSA algorithm. The improved SSA algorithm was then used to tune the model's parameters. The proposed model outperformed SVM, RF, BP, and PSO-BP models in metrics such as MAE, MAPE, RMSE, and R^2 for training and test sets. A summary of MH and ML-based studies is represented in Table 3.

Table 3. Hybrid MH-ML studies with ARO.

Study	Focus on	Role of ARO	Hybrid with
[80]	Dry density of soil stabilization blends	To tune the parameters of naïve a based model	GBO and Naïve Bayes
[81]	Compressive strength and flexural strength	To optimize the predictor model	HBA and CATBoost Regression (CAT)
[82]	Students' academic performance	To tune the parameter of the based model	JSO and Naive Bayes Classifier
[83]	Audit classification	To Feature extraction for use in SVM and ANN	SCSO and SVM-ANN
[84]	Detection accuracy of laser methane sensors	To improve the initialization of the sparrow population	SSA and BP neural network

4.1.3. Deep Learning-based studies

Another critical area where the ARO algorithm is used in hybrid form is studies based on DL-based models. In [85], the authors focus on emotional acknowledgment in the wild. Emotion identification in the wild is difficult due to head movement, facial deformation, and lighting fluctuation. Therefore, in this study, a deep learning-based model called Residual Neural Network (ResNet) is developed. The ResNet performs feature extraction and emotion

classification. In this regard, the ARO algorithm was employed to increase the reliability of this classifier. In this regard, AffectNet, AFEW, and RAF-DB datasets were used, and according to the results, the proposed hybrid model had a 5 to 8 percent higher accuracy rate than existing approaches. The [86] study has focused on providing and sustaining safe and high-quality products in the food supply chain. Therefore, a convolutional neural network (XAI-based Faster RCNN) model was developed to evaluate the contents of food items. Additionally, the ARO is implemented to register each user, assign a key, and secure the connections. This study used Blockchain (BC) technology on the security side. According to the results, the hybrid method is claimed to have an accuracy of over 99% with minimum processing time. [87] has addressed the issue of intrusion detection in intelligent healthcare. For this reason, the authors suggested an AI-based model based on the Hopfield Neural Network (HNN) and Deep Extreme Learning Machine (DELM). Besides, the ARO is applied to optimize the parameters of the HNN and DELM. This model, which leverages BC, classifies the presence of malicious activities in the healthcare system and helps in taking necessary actions to predict the disease. In this study, BC is used to secure the transmission of healthcare data. According to experimental results, it is claimed that the proposed method achieves superior performance compared to other approaches.

Another DL-based hybrid study has been considered for the stock price prediction issue [88]. This structure is a general approach that shows countries and companies' economic health and performance. An LSTM-based model has been developed to predict stock market prices based on multiple input and output time-step data. Also, the ARO is employed to tune the hyperparameters of the LSTM model. The DJIA index stocks were used as a data set, and the proposed model was compared with ANN and LSTM-GA-based methods. The proposed model was superior to others in MSE, MAE, MAPE, and R^2 parameters. Another similar study discussed stock price prediction and trading [89]. In this regard, LSTM and Wavelet-LSTM models have been proposed. The ARO is also used to optimize the hyperparameters of each model. It has been presented that better results were obtained when the ARO algorithm was adapted to these two models than the unadopted versions. The [90] focused on the early detection and classification of agricultural product diseases. A Deep Neural Network (DNN)-based model has been developed for more accurate detection and classification of images. This classifier will be used to convert the images from one domain to another in a paired manner. The ARO algorithm was used to tune the hyperparameters of the relevant model. The results were compared with two Generative Adversarial Network (GAN)-based methods, and the proposed model was claimed to perform better. The accuracy rate of the proposed model is claimed to be 99.7%.

In [91], the aim is to make a practical landslide susceptibility estimation. In this regard, a wide variety of ML and MH-based algorithms were used, focusing on a region of Vietnam. Finally, a decision-maker model was created. This model is based on the ARO. This algorithm enhances the model's accuracy, tunes the parameters, and finds their optimal values. According to the results, the AR-based model had a 0.95 AUC value, which is claimed to be better than models such as MPA and PSO. The [92] deals with user association problems in Ultra-Dense Networks (UDN). For this, Ambient Intelligence Exploration Multi-Delay Deep Deterministic Policy Gradient (AIEMDDPG) is suggested to solve data rate and latency problems in small and macro base stations of the Wireless Sensor Network (WSNs) [93]. Besides, the ARO is applied to improve the training efficiency of the proposed model. Lower latencies and higher data rates are tried to be achieved with this algorithm. In [94], a model based on a Deep Belief Network (DBN) for energy management of Air conditioning systems has been proposed. The authors have applied the ARO for optimizing the parameters of the DBN. A fitness function suitable for the problem has been defined in this regard. The R^2 value was improved from 0.83 to 0.93 with the proposed model. Moreover, improvement in accuracy has also been noted. This hybrid model outperforms similar models by optimizing energy consumption and costs in the respective systems. Deep learning-based ARO studies are summarized in Table 4.

Table 4. Hybrid DL studies with ARO.

Study	Focus on	Role of ARO	Hybrid with
[85]	Facial expression recognition	To choose the best value in feature selection	ResNet
[86]	Food supply chain management	To secure connection	CNN and BC
[87]	Intrusion detection in competent healthcare	To optimize parameters of HNN and DELM	HNN, DELM, and BC
[88]	Stock price prediction	To tune hyperparameters of the LSTM model	LSTM
[89]	Stock price forecasting and trading	To optimize the LSTM hyperparameters	LSTM and W-LSTM

[90]	Early detection and classification of agriculture products	To improve plant leaf disease classification by hyperparameter optimization	DNN
[91]	Landslide susceptibility prediction	To tune the unknown parameters	DNN
[92]	User association problems in ultra-dense network	To improve training efficiency	Deep reinforcement learning (AIEMDDPG)
[94]	Energy management of air conditioning systems	To optimize DBN	DBN

4.2. Variants of ARO

The MH approaches can be used to solve single or multi-objective problems. The binary variants evaluate issues such as condition monitoring and error diagnosis in complex systems within a completely discrete binary model. In multi-objective problems, unlike single ones, more than one optimum solution is presented. This section examines studies focused on these issues related to the ARO. In 8 studies published so far, appropriate fitness functions have been defined, mainly in distribution networks, food supply chains, microgrids, power systems, IoT, AC filters, Photovoltaic and electric vehicles areas, and the relevant algorithm has been transformed into a suitable state. These studies are summarized in this section.

4.2.1. Binary

A binary ARO [95] with four versions has been developed to feature selection with medical diagnosis data. These use S-shaped (AROS) and V-shaped (AROV) transfer functions. Moreover, in another variant, the exploration of ARO is improved by the Opposition-Based Learning (OBL) technique, and the levy flight strategy enhances the exploitation phase. In this model, a V-shaped structure named BGAROSLO was used. The last variant is based on the S-shaped model and the global-best concept (BGAROV). The analysis used 23 medical feature selection datasets and covid19 dataset, considering classification accuracy, specificity, sensitivity, fitness value, and the number of selected features. According to the results, the BGAROSLO approach appears superior to others.

4.2.2. Multi-objective optimization

The [96] study has focused on renewable energy sources such as solar PV, Wind Turbine (WT), and Distribution-Static synchronous Compensator (D-STATCOM) units, and the ARO algorithm was employed to locate and size relevant structures. This study generally intends to improve an electrical distribution network's performance and power quality (IEEE 33-bus). The ARO-based method obtained good results because it accelerated convergence and improved solution quality and comprehension. The [97] has focused on minimizing losses in network deployment. To achieve this goal, the Distribution Synchronous Static Compensator (D-STACTOM) can be used as a shunt compensator device in electrical distribution networks. The ARO algorithm allocates DGs and D-STATCOM, considering power, voltage, and cost parameters. This proposed algorithm is compared with two algorithms named Harris Hawks Optimization (HHO) [98] and Emperor Penguin Colony (EPS) [99] for optimal sizing and placement problems. According to the results obtained, it was claimed that the ARO-based algorithm exhibits better performance. The [100, 101] studies have focused on Load Frequency Control (LFC) in Isolated microgrids (IMG) with uncertain Renewable Energy Sources (RES). The two main methods for developing LFC schemes are adding new resources to IMG structures or designing new controller structures. However, another solution in designing this schema is improving the tuning procedures of the controller. Therefore, the authors of both studies proposed a new multi-objective model. In the first study, the tested IMG includes many sources, including diesel engine generators, battery energy storage technologies, fuel cells, and RES (such as photovoltaics and wind turbines). Therefore, in nonlinear states, the ARO algorithm was used to optimize the controller settings, considering various constraints of IMG resources. In another relative study, the ARO has been used to set controller parameters for various controlled sources in the IMG. On the other hand, the ARO is employed to tune Proportional-Integral (PI) and Proportional-Integral-Derivative (PID) controllers in both studies.

The [102] focused on restructuring the dynamic network and integrating temporary open points in the best way to overcome various uncertainties in active distribution networks with the increasing tendency towards renewable energies due to reasons such as global warming and increasing fuel prices. In this study, the location, size, and reconstruction of temporary open spots are obtained for the hourly load profile that includes the load penetration of the electric vehicle fleet and PV distributed generation. In this regard, a multi-objective function was defined, and

the ARO algorithm was used to solve this optimization problem. According to the results obtained from the simulations, the proposed approach offers a model with better technical and reliable features. The [103] focused on the efficient agricultural food supply chain, and in this context, a function based on a multi-objective closed-loop was defined by considering both forward and reverse product movements. The ARO and Simulated Annealing (SA) [104] algorithms were used hybrid to solve this optimization problem, and the results were compared with six multi-objective optimizers and three hybrid algorithms found in the literature. According to the analysis, the proposed algorithm provided better results. The [105] focuses on reducing data loss and minimizing energy consumption in Cognitive radio networks (CRNs). In this regard, a multi-objective function was defined to design an effective clustering structure, and the ARO algorithm was used to solve this NP-hard problem. As a result of the study, a routing protocol was proposed, and analyses were made on parameters such as remaining energy, packet loss rate, successfully sent packet rate, throughput, delay, and routing load. According to the results obtained, the proposed approach performs well. In [106], researchers focus on the design of AC filters, proposing an optimized Triple-Tuned Filters (TTFs) structure. This design reduces harmonic distortion in various operating modes and meets reactive power support criteria. The authors, who emphasized that the parameters should be optimized for the effective design of TTF, proposed a multi-objective ARO algorithm in this context. This function is used in an example system presented in IEEE STD-519, where harmonic, power quality, and loss parameters are considered. It was compared with the Ant Lion Optimizer (ALO) [107] and WOA to verify the effectiveness of the proposed design, and according to the results of the analysis, it was observed that the ARO-based model performed well. A summary of all variant studies on ARO is presented in Table 5.

Table 5. Variants of ARO-based studies.

Study	Focus on	Role of ARO	Variant
[95]	Medical diagnosis	To feature selection	Binary
[96]	Renewable energy source	To locate and size solar photovoltaic, wind turbine, and D-STATCOM units	MOP
[97]	Electric distribution Networks	To optimal allocation problem of DGs and DSTATCOM, considering power, voltage, and cost parameters	MOP
[100]	Isolated microgrids	To tune PI and PID controllers	MOP
[101]	Isolated microgrids	To tune PI and PID controllers	MOP
[102]	IEEE 33-bus electrical distribution networks	To solve the multivariable optimization problem	MOP
[103]	Agri-food supply chain	To solve a multi-objective closed-loop supply chain	MOP
[105]	Cognitive radio network	To optimal clustering and routing	MOP
[106]	Design of AC filters and TTF	To optimize the filter parameters to mitigate harmonics, improve power quality, and minimize power losses	MOP

4.3. Improved class

This section examines studies on improving, changing, or developing the ARO algorithm with various mechanisms. In the literature, improvements have been made to this algorithm, like many MH-based algorithms [108, 109]. Researchers have pointed out the limitations of the ARO algorithm in solving different problems, such as slow convergence, optimal local trap, and exploration and exploitation phases that cannot cope with extensive dimensions. Therefore, various approaches and strategies, such as levy flight, opposition-based learning, and chaotic, have been used to improve the limitations of ARO. All the methods used benefited from the 25 techniques presented in Figure 9. In studies in the literature, multiple strategies have been used from time to time to improve, enhance, or modify the ARO algorithm. For this reason, we define subgroups of studies according to the number of techniques used and summarize the improved ARO algorithm according to the strategy used in each.

4.3.1. Studies based on a single strategy

In [110], an improved ARO algorithm based on the *Levy Flight (LF) strategy* has been proposed to achieve energy and distance-aware routing mechanisms in WSNs. The LF was used to prevent ARO from falling into the local solution and to increase convergence accuracy. The modified ARO algorithm is applied to find optimal values for uncertain parameters of the network. Analyzes were made on network throughput, load balance, and energy efficiency, and according to the results, the proposed method achieved better outcomes. In the [111] study, another modified ARO algorithm was proposed for MPPT. Because Solar Photo-Voltaic (SPV) systems may experience

power losses due to various environmental conditions, in this version of ARO, the *LF strategy* was used in the random hiding phase to improve the original algorithm's flexibility, convergence, population diversity, and exploitation phase. The proposed algorithm was compared with PSO, ARO, and SSA algorithms, and the results were claimed to be promising. The [112] introduced a Nonlinear Chaotic ARO (NCARO) algorithm for task scheduling problems in a fog computing environment. In this study, the *chaotic strategy* was used to improve the exploration behavior of the standard ARO. This study used ten chaotic maps focused on service time, cost, and energy consumption parameters. As a result of the analysis, the enhanced ARO algorithm based on the Sinusoidal map exhibited the best performance. The [113] study focused on solving various problems in the field of Radial Distribution Networks (RDN), and in this regard, a modified version of the ARO was presented. This algorithm helps improve the Voltage Stability Index (VSI) and optimal placement and size of Distributed Generation (DG). In this study, it is planned that the ARO algorithm will perform a more effective search and obtain a superior result using the *Opposition-Based Learning (OBL) strategy*. The analysis has been tested on IEEE 33-bus and 118-bus systems, and the results demonstrate that the presented method exhibits better performance than others.

In the [114], the *Diagonal Linear Uniform (DLU) strategy* was used instead of the random start of the ARO algorithm. The main reason is to improve the algorithm's overall performance and convergence behavior. The problem the authors are based on is realizing the optimum design of lattice structures. The algorithm has been implemented for this problem, and famous benchmark test functions are better than the original ARO. The [115] proposed a Modified ARO (MARO) to improve standard ARO's local optima and low diversity issues. This algorithm combined the ARO with the *Dimension Learning-Based Hunting (DLBH) strategy*. A PV, WT, and Biomass hybrid system was used as the test problem and system. The analysis was compared with the original ARO, Dandelion Optimizer (DO), and Driving Training-Based Optimization (DTOB). The results indicate that the proposed algorithm is promising. The [116] has focused on the heat and power economic dispatch problem. In this regard, the ARO was developed with an *Adaptive Fitness Distance Balance (AFDB) strategy* to solve the problem of getting caught in local solution traps in the search space, as well as to improve searchability and establish a good balance between the exploration and exploitation process. In [117], a new ARO algorithm (Leader ARO) version was defined for designing the optimal hybrid microgrid system. It uses a *leader-based mutation-selection (LMS) strategy to improve the ARO's exploration performance*. As a result of this study, costs were analyzed based on microgrid configuration using PV/wind/diesel/battery, and ARO and other MH algorithms were used in the literature. According to the results, the proposed improved version provides good value.

In [118], the aim was to optimally size and minimize the life cycle cost of a PV pumping system with a Water Storage Tank (WST), considering the limitation of the possibility of intermittent water. In this regard, a *Non-Linear Inertial Weight Reduction (NLIWR) strategy* was used to overcome the premature convergence of the standard algorithm by defining a new version of the ARO algorithm. The proposed method was compared with ARO and PSO and according to the results, the enhanced version supplied customers' water demands with higher costs and lower reliability. In [119], a Modified ARO (MARO) algorithm was proposed to solve the energy management problem in microgrids. This algorithm serves as optimum planning for microgrid resources. For this, improvement was made with *Quantum mechanics based on the Monte Carlo method*. The main aim is to maximize the benefit of the MG operator and minimize the daily operating cost. The MARO improves the original ARO algorithm's exploration phase and population diversity. In [120], an improved version of ARO has been proposed to calculate the best capacity of hub energy equipment in line with hub energy management. The primary focus of this optimization problem is finding optimal values for the uncertainty of solar and wind energy sources. In the proposed algorithm, the ARO is improved by *a non-linear dynamic weight inertia (NLDWI) strategy to balance the exploration and exploitation phases and increase the algorithm's local search capacity* against premature convergence. Summaries of studies that improved the ARO algorithm with a single strategy or method are reported in Table 6, along with their features.

Table 6. Improved ARO-based studies by a single strategy.

Study	Focus	Role of ARO	Improved by	Advantages
[110]	Route in WMNs	To find optimal values for uncertain parameters	Levy flight strategy	Prevent ARO from falling into the local solution and increase convergence accuracy

[111]	Solar PV system	To MPPT	Levy flight strategy	Improve the flexibility, convergence, population diversity, and exploitation behavior
[112]	Task scheduling in a fog computing environment	To find optimal values in service time, cost, and energy consumption parameters	Chaotic strategy	Improving the exploration phase to avoid local optima trap
[113]	Radial distribution network	To improve the voltage stability index and the optimal placement and size of DG	Opposition-based learning	Provides more effective performance in search
[114]	Design of truss structures	To find optimal values in truss structure design	Brownian random motion strategy	Enhance the overall performance and convergence behavior
[115]	PV, WT, Biomass hybrid system	To find optimal values of the designed hybrid system	Dimension learning hunting (DLH) strategy	improving local optimum trap and low diversity problems
[116]	Economic dispatch problem	To find optimal values	Adaptive fitness-distance strategy	Improving balancing in exploration and exploitation behavior by providing diversity
[117]	Hybrid microgrid system	To find optimal values	Leader-based mutation selection technique	Improving the exploration phase
[118]	PV pumping system	To the optimal selection of the inertia weight	Nonlinear inertia weight reduction strategy	Overcome the premature convergence
[119]	Microgrid energy management	To optimal scheduling for MG resources	Quantum mechanics based on the Monte Carlo method	Improving the exploration phase and population diversity
[120]	Hub energy management	To calculate the best capacity of hub energy equipment	Non-linear dynamic weight inertia	Ensure balance between phases and increase the local search capacity

4.3.2. Studies based on multi-strategy

This section examines studies that improve or modify the ARO algorithm with multiple approaches and strategies. The [121] has focused on the efficiency of wireless communication systems. A Reconfigurable Smart Surface (RIS) is used to achieve this goal. However, the ARO algorithm was improved and applied since it was difficult to determine the amount and locations of RIS elements. In this new version, the exploration phase of ARO with the *Collaborative Search Operator (CSO) strategy* and the exploitation phase with the *Adaptive Time Function (ATF)* have been improved. The results of the proposed algorithm were compared with Growth Optimizer (GO) [122], Artificial Ecosystem Optimizer (AEO) [123], PSO, and ARO, and good performance was observed. The [124] identifies models with Infinite Impulse Response (IIR) in signal processing and system identification. In this regard, an improved ARO algorithm is proposed to determine the coefficients of IIR for estimating unknown plants. The Adaptive Local Search (ALS) mechanism and Experience-based Perturbed Learning (EPL) strategy improve the ARO algorithm. These additions address the loss of accuracy during iterations and improve exploitability. In [125], these two strategies (ALS and EPL) were adapted to address the above problems of ARO and used to solve a different issue (automatic voltage regulators). The results demonstrate that this version of ARO also successfully solves this problem. The problem analyzed in [126] is weak fault feature extraction and the enhancement of Autonomous Underwater Vehicle (AUV) thrusters based on ARO and Variational Mode Decomposition (VMD). In this regard, the ARO algorithm is improved based on *chaotic maps* and *Gaussian mutation* strategies. This enhanced ARO is generally used to optimize the parameters of VMD. The proposed method was implemented on the AUV prototype, and the results were claimed to be efficient and effective.

The [127] study improves the diversity and dynamics of the population of ARO by *LF strategy* and escapes from local optimal by *Selective Opposition (SO) strategy*. The proposed ARO is tested over benchmark global optimization problems and compared with various MH algorithms. Experimental results show that the proposed algorithm is good at complex optimization problems through different performance metrics. In [128], a new ARO algorithm is proposed for optimal scheduling of various energy resources in microgrids. In this regard, the ARO has been improved with

LF and *chaos* strategies. The first technique enhances the exploration and escapes from local optima, and the second strategy is applied to enhance the exploitation phase. The [129] study focuses on optimal power flow problems inspired by the ARO algorithm. In the first stage, the ARO is improved by *Roulette Fitness-Distance Balance (RFDB)* and *dynamic Fitness-Distance Balance (dFDB)* strategies. In this regard, the optimal allocation of FACTS devices (SVC, TCSC, and TCPS) is performed with dFDBARO to optimal power flow in the RESs (solar and wind). This study considered the case study as an IEEE 30-bus power network. In the analysis phase, a comparison was made with other powerful algorithms (Equilibrium Optimizer (EO), Hunger Games Search (HGS), Lightning Search Algorithm (LSA), and Chaos Game Optimization (CGO)). The results obtained with this new algorithm have been claimed to be promising. This section's last study has focused on the ball NURBS curves in geometric modeling [130]. This paper proposes an enhanced ARO algorithm to accomplish the optimal multi-degree reduction of the ball curves. The ARO algorithm is improved by *mutation*, *adaptive group*, and *Elite-feedback* strategies. This new version of ARO helps increase population diversity and the ability to move outside local minima. Summaries of studies that utilized more than one strategy or method in improving the ARO algorithm, along with their advantages, are presented in Table 7.

Table 7. Improved ARO-based studies by multi-strategy.

Study	Focus	Role of ARO	Improved by	Advantages
[121]	RIS multiuser wireless communication systems	To minimize the number of RISs	Collaborative searching operator and adaptive time function	It provides more balanced transitions between phases
[124]	Infinite impulse response system	To identify the coefficients of IIR for estimating the unknown plants	Adaptive local search (ALS) mechanism and experience-based perturbed learning (EPL) strategy	Address the loss of accuracy during iterations and improve exploitability.
[125]	Automatic voltage regulators	To find optimal values	ALS and EPL	Address the loss of accuracy during iterations and improve exploitability.
[126]	Weak fault feature extraction and AUV	To long-running times and to optimize the parameters of VMD Fix the distortion problem in the fault feature extraction effect	Chaotic mapping and Gaussian mutation	Improve population diversity and exploration phase
[127]	Global optimization	To find optimal values	Levy flight and selective opposition	Improve the diversity and dynamics of the population and achieve local optimal
[128]	Optimal scheduling algorithm	To find optimal values	Levy's flight and chaos	Improve the exploration and escape from local optima by Levy flight; Improve the exploitation based on chaotic maps
[129]	Power flow problem	To find optimal values	Roulette fitness-distance balance (RFDB) and dynamic fitness-distance balance (dFDB) strategies.	Improving exploration phase
[130]	Ball NURBS curves in geometric modeling	To accomplish the optimal multi-degree reduction of the ball curves	Mutation, adaptive group, and Elite-feedback strategies	Population diversity and enhance its capability of jumping out of the local minima

Apart from these studies, ARO improvements are associated with ML and DL-based models in some studies. Therefore, these studies are examined separately below. In [131], the focus was on hydraulic unit vibration states, and an SVM-based identification model was developed for this. In this regard, an improved ARO (IARO) algorithm is proposed to optimize the parameters of the SVM multi-classifier. *The adaptive Weight (AW) strategy* improved the ARO algorithm's convergence accuracy, convergence speed, and local optimum avoidance characteristics. In

particular, this technique positively affects the optimization ability of the algorithm in terms of convergence speed. This model was compared with ARO-SVM, PSO-SVM, ASO-SVM, and WOA-SVM models and showed better results. The [132] study focuses on traffic flow monitoring in Intelligent Transportation Systems (ITS). In this regard, LSTM and BiGRU-based predictor model is suggested. The authors also proposed an improved ARO algorithm for optimal feature selection and to obtain an accurate model. For this reason, they use the *Adaptive Weight (AW) strategy*. Simulation results show that the ARO-based model is more effective and accrues with less RMSE value than other approaches. The [133] study proposed an LSTM-based model for predicting the stock price market. The authors improved the ARO algorithm based on the *Experience-based Perturbed Learning (EPL) strategy* to tune hyperparameters of the LSTM. In this study, ARO's location update system in the discovery phase claimed that it caused the invasive diversity model. The EPL technique was included in the algorithm as a solution. This strategy is used to find a lot of promising areas within the suitable search area, which increases the exploration phase. Since arrhythmia classification is difficult under normal conditions due to the overfitting problem caused by the high-dimensional feature space of the ECG signal, a CNN-based model was developed in the study [134]. In this regard, the *Selective Opposition (SO) strategy* was used to improve the ARO algorithm in the exploitation phase. This modified ARO algorithm selects the most suitable feature subset from the overall features to avoid the overfitting problem. The SO helps obtain the optimal solution in the development phase by changing the proximity size of various rabbits. In the [135] study, an RL-based model was developed to create synergistic registration of CT-MRI brain and retina images. In this regard, a hybrid structure of the ARO algorithm was designed, and ARO was also improved in the study. It leveraged the *Q-learning algorithm, an adaptive strategy-selection mechanism for improving exploitation*, and the *LF strategy* for the exploration phase. The results of both registration experiments claimed that the RL-mARO model exhibited high robustness and registration accuracy. The [136] study focused on skin cancer prediction based on images. For this purpose, a DL-based model has been developed, and an improved ARO algorithm has been proposed for feature selection. In this new version of ARO, *Gaussian mutation* and *crossover operator* are used. These strategies are used to prevent diversity loss during the search process and to ignore the unimportant features from those features extracted. The [137] develops a KNN-based model for network attack detection. In this regard, a new version of the ARO algorithm is proposed and used for optimal feature selection for the relevant classifier. In this context, ARO has been improved with four different approaches. 1) Uses *Bottleneck Dolphin Optimizer (BDO)* [138] to improve global exploration and fast search of the ARO. 2) The algorithm tries to achieve more balance between phases with the *adaptive switching mechanism*. 3) Better local usage is achieved with the *LF strategy*. 4) *Dynamic Lens Imaging Learning (DLIL) strategy* is used to better escape from the local optimum. In the [139] study, a Convolutional Bidirectional LSTM (CBiLSTM) based model was developed to increase efficiency in PV energy production and optimize and monitor its consumption. Additionally, the ARO algorithm has been improved to fine-tune this model. In the new version of ARO, an *opposition-based learning strategy* is used to increase diversity, an *LF strategy* to escape local optimum, and a *Sine-cosine strategy* to be balanced between phases. A summary of the studies at this stage is presented in Table 8.

Table 8. Improving ARO-based studies in the interaction of ML and DL-based methods.

Study	Focus	Role of ARO	Improved by	Advantages
[131]	vibration states of hydraulic units	To optimize the parameters of the SVM multi-classifier	Adaptive Inertia Weight Strategy	Improves convergence accuracy, convergence speed, and local optima avoidance
[132]	Transportation system	To feature selection	Adaptive weight	Improving convergence speed
[133]	Stock market price prediction	To tune the parameters of LSTM	Experience-based perturbed learning (EPL)	Improving the exploration behavior
[134]	Electrocardiogram IoMT	To optimal feature subset selection	Selective Opposition (SO) strategy	Improving the exploitation phase to obtain the best solutions
[135]	CT-MRI brain and retinal images	To Fine-tuning specific hyperparameters	Q-learning as an adaptive strategy-selection mechanism and Levy flight	Trying to guarantee a global optimum, especially in the last iterations

[136]	Skin cancer prediction based on images	To feature selection	Gaussian mutation and crossover operator	Prevent diversity loss during the search process
[137]	Network intrusion detection	To feature selection for the KNN classifier	Bottlenose dolphin optimizer (BDO), adaptive switching The mechanism, Levy flight strategy, and dynamic lens-imaging learning strategy	Improving global exploration and fast search, more balancing between phases, better local exploitation, escaping from the local optimum
[139]	Photovoltaic	To fine-tune hyperparameters	Opposition-based learning, sine cosine, and Levy flight strategy	Increasing diversity, escaping local optima, and more balancing in phases

4.4. Adapted class

This section discusses studies that adapt the standard ARO algorithm to solve problems in various science branches without any changes. In line with the specific features of the ARO algorithm, many researchers prefer it for solving multiple issues. These optimization problems are mainly in engineering disciplines and consist of prediction, feature selection, complex high-dimensional optimization, energy management, scheduling, fine-tuning, and clustering. As presented in Figure 10, 33% of the total citations belong to this category. This category, which has the highest percentage, implies that ARO is fundamentally effective and robust. The researchers directly benefited from the ARO algorithm in solving various problems addressed in their studies, and fitness functions compatible with the problem were defined to adapt the relevant algorithm. In examining the studies in this section, the issues focused are presented under four basic headings according to the primary science branch and field, without dividing into too many subclasses. Detailed information on the studies included in this category is also presented in Table 9. It is worth emphasizing that the studies evaluated in the previous three categories also focus on solving complex problems in different application areas. Application areas of all categories are presented in Figure 12.

Signal/Image processing

The [140] proposed an algorithm based on a Multi-bit Local Neighborhood Difference Pattern (MLNDP) optimized for robust and accurate detection of West syndrome. The ARO algorithm was used to select the optimal parameters. In another signal processing-based study [141], the ARO algorithm, which focused on the kurtosis product of the noise ratio, was used to find the best filters. In [142], the authors focused on developing an IIR filter-based Digital Fractional Order Differentiator (DFOD) and, to help this, used the ARO to find the optimal value for the coefficient of the IIR filter. Other similar studies are found in the relevant references [143, 144].

Power and energy systems

In [145], PV and battery power efficiency were considered, and ARO was adapted to tune the Fractional Proportional Controller (FOPI) in these systems. [146] has been focused on electricity suppliers such as PV, wind turbines, and fuel cells, and in this regard, the ARO is applied to reduce the cost of electricity. The [147] study considered PV and Thermo-Electric Generation (TEG) systems, and the ARO algorithm was used to find the optimal irradiation distribution. The [148] has adapted the ARO to minimize customer energy costs by considering wind generators and storage systems. In [149], automatic production control is discussed with a focus on disturbances such as thermal, geothermal, realistic dish-Stirling solar thermal, and sensitive wind turbine systems. The main problem is accessing the controller's properties; the ARO algorithm has been adapted in this context. In [150, 151], it aims to improve the frequency performance of power systems by manipulating reactive power and wind-Integrated Power Systems in STATCOM. The authors utilized ARO to find and tune the parameters of the PD-PIDA controller. In [152], the ARO has been employed to find optimum values of uncertain parameters in the Proton Exchange Membrane Fuel Cell (PEMFC). In [153], the ARO is adapted to best control MGs. For this reason, the relevant algorithm is used to fine-tune the control parameters. [154] has been focused on MPPT controllers in PV systems to get the maximum power available. The ARO is used to extract maximum power from the photovoltaic module. In [155], researchers are trying to solve the Economic Load Dispatch Problem (ELDP) of a three-generating unit system in Thermal power plants that includes valve-point loading effects using the ARO.

In [156], it is emphasized that power systems' most significant problem is ensuring harmony between load demand and generation demand. The authors focus on Multi-Area Power Systems (MAPS) and propose an optimal PID controller using ARO for LFC in MAPS of two non-reheat thermal systems. In the [157] study, the optimal location and rating of SPV Systems and passive power filters (PPFs) in the IEEE 33-bus Radial Distribution Network (RDN) field were discussed. The ARO algorithm was used to achieve this goal. The [158] study was focused on power quality problem in PV systems. The ARO is applied to obtain the best tune of the PI controller. In [159], the ARO algorithm was used to find optimum values for unknown parameters in the SPV systems. Similarly, the [160] focuses on solar PV systems. This study utilized ARO to find the optimal parameter identification of the PV cell model. The [161] study focused on Automatic Voltage Regulator (AVR) systems. It explained that these systems in the production station use the PID controller to adjust the abnormal voltage caused by transient conditions. The ARO algorithm was used to tune these controller devices optimally. The [162] study explains that defining suitable electrical models and determining the maximum power point with Perovskite Solar Cells (PSCs) is possible, but dynamic variables must be controlled. In this context, ARO was utilized to optimize the parameters of the PSC dynamic model. The study [163] investigated the control of aircraft pitch angles as exceptionally sensitive due to flight's nonlinear and uncertain dynamics. In this regard, ARO is used to tune the PID controller with a filtering mechanism in this control system. Modern electrical networks were discussed in another study focused on energy and power systems [164]. Unbalanced load must be managed because inertia constantly decreases and is sensitive to load/production change. In this regard, an improved LFC method is proposed. In this design model, the ARO algorithm was used to optimize all adjustable parameters simultaneously. The [165] study discussed distribution system losses, voltage profiles, and voltage stability problems. In this regard, the ARO algorithm was used for optimal positioning and sizing of distributed generation.

Smart and autonomous systems

The [166] study focused on the problem of effective task scheduling for efficient resource consumption in IoT systems, and the ARO algorithm was used to solve this problem. The [167] uses the ARO to perform pattern recognition and navigation in Automatic Guided Vehicles (AGV). In the [168] study, the ARO was used to find path errors and angles by focusing on the kinematic issue in robotics. The [169, 170] studies focused on developing efficient routing protocols in WSN and IoT systems. In this regard, both studies utilized the ARO algorithm. In the first one, the aim is to choose the optimum Cluster Head (CH), and the other research aims to select the best relay by balancing the EC of the forwarder and the packet delivery rate. In another study [171], the ARO algorithm is applied to select the optimal CH element for energy efficiency in WSNs.

Other applications

The [172] study focused on a problem in sports from a different perspective. This study tries to solve the problem of football players' lack of exercise in various subjects for effective football teaching and training. In this regard, a graph sample and aggregate attention network (GSAAN)-based model was proposed. GSAAN filtering was improved by finding optimum weight parameters with the help of the ARO. In [173], the beam optimization problem was discussed regarding the inverse problem, one of the famous optimization problems in mechanical engineering. For this, the ARO algorithm was used to find the optimum values. In [174], the focus is on determining the ideal parameters for resistance spot welding, a topic in the manufacturing sector, using the ARO algorithm. In [175], the authors proposed a practical framework based on the accurate equivalent Single Degree of Freedom (SDF) system to highlight the efficient and precise evaluation of structural collapse capacity in earthquake engineering. This system used ARO to calibrate the Cyclic Static Pushover (CSP) curves of a more complex Multi-Degree of Freedom (MDF) system. In [176], the Variable Exponential Nonlinear Muskingum Model (VEP-NMM) is recommended to increase flood routing accuracy for efficient use of water resources. The ARO was used to solve this model's parameter optimization problem.

Table 9. ARO-based studies in adapted class.

Study	Focus	Role of ARO	Category
[140]	Multichannel EEG	To optimal parameter selection	Signal/Image processing
[141]	NPKPD	To find the best filters	Signal/Image processing
[142]	Digital fractional order differentiator (DFOD)	To help in developing IIR filter-based DFOD (find the optimal value for the coefficient of IIR filter)	Signal/Image processing

[145]	PV and battery power	To tune FOPI	Power and energy system
[146]	Electricity supply (PV, wind turbine, fuel cell)	To reduce cost	Power and energy system
[147]	PV-TEG systems	To find optimal irradiation distribution	Power and energy system
[148]	Wind generator and storage systems	To minimize energy costs for customers	Power and energy system
[149]	Automatic Generation Control	To attain the controller's features	Power and energy system
[150]	STATCOM	To find-tune the PD-PIDA controller's parameters	Power and energy system
[152]	Proton exchange membrane fuel cell (PEMFC)	To find optimum values of uncertain parameters	Power and energy system
[153]	Microgrid system	To fine-tune the control parameters	Power and energy system
[154]	MPPT controller in PV system	To extract maximum power from the photo voltaic module	Power and energy system
[155]	Economic Load Dispatch Problem	To solve the ELDP of a three-generating unit system that includes valve-point loading effects	Power and energy system
[156]	Multi-Area Power Systems	To optimize PID controller settings for the LFC problems	Power and energy system
[157]	IEEE 33-bus Radial Distribution Network (RDN)	To find the optimal location and rating of solar photovoltaic systems and passive power filters	Power and energy system
[158]	Power quality problem in Photovoltaic (PV) systems	To obtain the best tune of the PI controller	Power and energy system
[159]	Solar Photovoltaic systems	To determine the unknown parameters of PV models	Power and energy system
[160]	solar photovoltaic systems	To find optimal parameters identification of the PV cell model	Power and energy system
[161]	Automatic Voltage Regulator (AVR) system	To tune the PID controller	Power and energy system
[162]	Perovskite solar cells (PSC)	To optimize the parameters of the PSC dynamic model	Power and energy system
[163]	Aircraft pitch angles	To tune a PID controller with a filtering mechanism (PID-F)	Power and energy system
[164]	Modern power grids	Finds all the tunable parameters	Power and energy system
[165]	Radial distribution system (RDS)	To optimal DG Allocation Problem	Power and energy system
[166]	IoT	To efficient task scheduling	Smart and autonomous systems
[167]	AGV	To pattern recognition and navigation	Smart and autonomous systems
[168]	Kinematic	To find path errors and angles	Smart and autonomous systems
[169]	Routing technique in IoT	To select optimal CH	Smart and autonomous systems
[170]	Routing technique in IoT	To pick the best relay by balancing the forwarder's EC and packet delivery ratio (PDR)	Smart and autonomous systems
[171]	Cluster head selection in WSN	To optimal cluster formation	Smart and autonomous systems
[172]	Effective soccer teaching and training (football)	To improve the filtering of GSAAN by finding optimal weight parameters	Other (Sport)
[173]	Inverse problem (beam)	To find optimal values	Other (Mechanical engineering)
[174]	Resistance spot welding	To determine ideal parameters	Other (Manufacturing sector)
[175]	Degree of freedom systems	To find optimal values for hysteretic parameters of MDOF and ESDOF systems	Other (Earthquake engineering)

[176]	Flood routing	To solve the parameter optimization problem of the Muskingum model	Other (Water resources management)
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5. Applications of ARO-based studies

In this section, in addition to the classifications, ARO-based applications in the literature are also presented in summary form in Figure 11. Accordingly, related studies have mainly used ARO to solve specific real-world problems or have been tested in classical engineering or well-known benchmark test functions. These problems and their application areas are primarily included in the adapted class, but various issues are also considered in the other three classes. In addition to this information, the percentage distribution of multiple applications based on the ARO algorithm is presented in Figure 12.

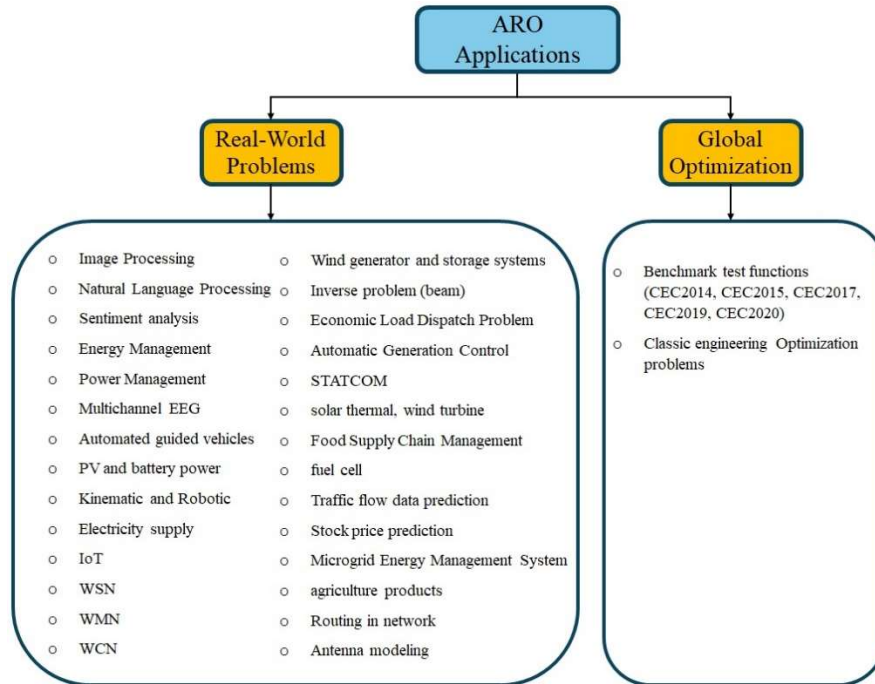


Figure 11. Applications of ARO-based studies.

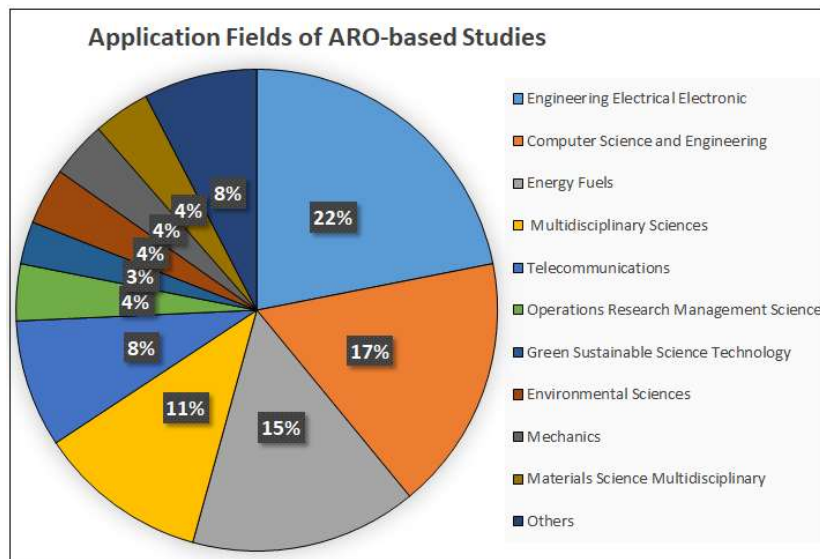


Figure 12. Percentage distribution of optimization problems based on ARO algorithm.

6. The Behavior Analysis of ARO

In this section, we present the results of examining the efficiency and performance of ARO on 20 classical benchmark test functions from the CEC category (Appendix 1) as an example from numerical analysis. They are the classical

functions many researchers utilize [2, 40, 48]. In this regard, a comparison is made with five well-known algorithms (SCSO [78], MPA [177], WOA [54], SSA [79], ASO [33]). The hyperparameters of all algorithms are set, as shown in Table 10. The simulation was presented using MATLAB on the Core i7-11800U 2.3 processor with 16GB of RAM computer. In the evaluation, the population size, maximum iteration, and number of independent runs for all algorithms were considered 50, 1000, and 30, respectively.

Table 10. Simulation parameter for each algorithm.

Algorithm	Parameter	Value
ARO	N1	[0, 1]
	N2	[0, 1]
SCSO	Sensitivity range (r_G)	[2,0]
	Phases control range (R)	[-2rG, 2rG]
MPA	α	3
	β	3
	γ	1.5
WOA	a	[2,0]
	A	[2, 0]
	C	2.rand (0,1)
	l	[-1,1]
	b	1
SSA	Initial speed (v_0)	0
	Leader position update probability	0.5
ASO	Depth weight	50
	Multiplier weight	0.2

The first 7 of the 20 test functions are unimodal, the second 6 are multimodal, and the last six are fixed-dimension multimodal. The variety of the selected functions has been increased to improve the accuracy of both the results obtained and the analysis performed. The unimodal ones have only one global optimum and no local optimum. Multimodal functions have multiple local optimum but only one global optimum. This benchmark function helps evaluate the performance of optimization algorithms in the exploration and exploitation stages. In fixed-size multimodal functions, the size of each benchmark function is static, and the dimension number cannot be adjusted, while in multimodal functions, it can be adjusted [78]. The analysis results are presented in Table 11. In unimodal test functions, the ARO achieved the best performance in 6 functions, achieving an 86% success rate. When standard divisions are analyzed, it is understood that the convergence performance is similar. In multimodal, the ARO has the best performance in 4 functions, has a success rate of 67%, and ranks first with the SCSO algorithm. It is understood that it also ranks first in convergence speed. The algorithms following this algorithm are ASO, MPA, WOA, and SSA. Although almost all algorithms indicate similar performance in the last category of test functions, the ARO algorithm maintains its first place. Consequently, when the results were analyzed, the ARO algorithm performed well on all test functions except for four functions (F7, F12, and F13). This means 85% superiority and success rate. In these analyses, SCSO ranked second, ASO ranked third, and MPA algorithm ranked fourth. The last two places belong to WOA and SSA algorithms, respectively. In ARO's original paper and modified versions, simulations were performed on both test functions and real-world optimization problems, and various analyses indicated that this algorithm performed well. In general, the ARO algorithm, like other MH algorithms, has its shortcomings or weaknesses in addition to its advantages. The strengths properties can be summarized below. Its transition between the exploration and exploitation phases is balanced. It maintains stability in the presence of disturbances. It has a simple structure and ease of adaptation. It can quickly find the most suitable solutions for time-sensitive applications.

It can adapt to various problems with proper parameter tuning. It scales effectively to significant optimization problems [65]. The weaknesses of ARO can be listed as slow convergence, probability of falling into local optima, and lower population diversity. It should be emphasized that, like every algorithm, eliminating the shortcomings in ARO is a challenge, while further improving the strengths can be a subject of future work and research. Moreover, Table 12 presents a summary comparison of the main features of ARO with other optimization algorithms used for benchmarking in this section.

Table 11. Simulation result for each algorithm.

Function		ARO	SCSO	MPA	WOA	SSA	ASO
F1	Avg	3.93E-124	2.42E-97	1.132E-49	3.26E-71	3.08E-07	5.07E-22
	Std	6.29E-124	7.22E-97	1.929E-49	1.78E-70	7.18E-07	2.31E-22
F2	Avg	2.49E-73	1.16E-52	7.952E-28	9.39E-52	1.71E+00	1.24E-10
	Std	1.29E-73	2.55E-52	1.076E-27	3.56E-51	1.39E+00	1.82E-10
F3	Avg	1.64E-99	7.84E-81	2.511E-11	4.04E+04	1.47E+03	2.30E+02
	Std	2.44E-98	3.49E-80	1.083E-10	1.46E+04	8.31E+02	9.81E+01
F4	Avg	4.01E-55	4.57E-46	2.092E-19	4.60E+01	1.11E+01	2.34E-09
	Std	2.11E-55	9.98E-46	1.268E-19	2.90E+01	3.93E+00	2.25E-09
F5	Avg	4.80E-03	2.49E+01	2.359E+01	2.79E+01	2.47E+02	2.51E+01
	Std	5.02E-03	6.70E-01	4.947E-01	3.84E-01	5.22E+02	0.2338
F6	Avg	0.00E+00	2.15E+00	1.656E-09	4.88E-01	1.75E-07	0.00E+00
	Std	0.00E+00	4.47E-01	7.388E-10	3.17E-01	3.53E-07	0.00E+00
F7	Avg	1.90E-04	1.51E-04	5.805E-04	4.34E-03	1.59E-01	1.75E-02
	Std	1.55E-04	1.33E-04	4.771E-04	4.14E-03	5.53E-02	1.09E-02
F8	Avg	-1.01E+04	-1.01E+04	-9.622E+03	-6.98E+03	-7.39E+03	-8.4E+04
	Std	2.11E+01	1.70E+03	4.432E+02	7.03E+02	7.32E+02	6.32E+01
F9	Avg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	5.28E+01	2.5.6E+01
	Std	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.08E+01	6.08E+00
F10	Avg	8.77E-16	8.77E-16	3.642E-15	4.09E-15	2.46E+00	1.43E-11
	Std	0.00E+00	0.00E+00	1.084E-15	2.35E-15	6.88E-01	2.82E-11
F11	Avg	0.00E+00	0.00E+00	0.00E+00	1.49E-02	1.49E-02	0.00E+00
	Std	0.00E+00	0.00E+00	0.00E+00	4.65E-02	1.22E-02	0.00E+00
F12	Avg	3.56E-08	1.25E-01	1.705E-10	2.89E-02	7.34E+00	7.08E-24
	Std	1.22E-08	5.41E-02	9.113E-11	2.75E-02	2.59E+00	5.91E-24
F13	Avg	3.62E-07	1.99E+00	1.397E-03	5.66E-01	1.68E+01	4.76E-23
	Std	6.89E-06	2.51E-01	3.440E-03	2.72E-01	1.74E+01	6.03E-23
F14	Avg	9.980E-01	1.73E+00	9.980E-01	3.48E+00	1.23E+00	9.98E-01
	Std	0.00E+00	9.00E-01	5.831E-17	3.49E+00	5.01E-01	9.22E-12
F15	Avg	3.07E-04	4.82E-04	3.07E-04	8.07E-04	2.94E-03	8.02E-04
	Std	0.00E+00	3.22E-04	2.29E-19	6.06E-04	5.92E-03	1.65E-04
F16	Avg	1.03163	-1.03163	-1.03163	-1.0316	-1.0316	-1.0316
	Std	0.00E+00	1.15E-09	6.253E-16	1.99E-09	4.45E-14	6.72E-16
F17	Avg	0.398	0.398	0.398	0.398	0.398	0.398
	Std	0.00E+00	1.55E-07	0.00E+00	2.96E-05	0.00E+00	0.00E+00
F18	Avg	3	3	3	3	3	3
	Std	0.00E+00	1.67E-05	1.131E-15	1.59E-04	3.48E-13	2.21E-15
F19	Avg	-3.86	-3.86	-3.86	-3.86	-3.86	-3.86
	Std	0.00E+00	1.49E-03	2.710E-15	4.04E-03	2.47E-11	0.00E+00
F20	Avg	-3.32	-3.25	-3.22	-3.23	-3.22	-3.32
	Std	0.00E+00	9.95E-02	1.571E-15	1.41E-01	5.62E-02	2.71E-15

The best values are remarked on in bold

Table 12. Comparison between used algorithms.

	Hyperparameters number	General Complexity	Computational running time	Strength	Weaknesses
ARO	2	$O(T_{nd}+T_n+n)\cong O(n^2)$	4.2020 seconds	Stability, flexibility, easy implementation,	Slow convergence, Probability of

				no derivation mechanism, good performance in finding global optima, balance in exploration and exploitation phases switching, can quickly find the most suitable solutions for time-sensitive applications.	falling into local optima, and lower population diversity
SCSO	2	$T(n) = (n + 1) + (m + 1) + c(n \times m) + O(1) \cong O(n^2)$	4.8910 seconds	Easy implementation, balancing between phases	Population diversity
MPA	3	$O(T_{nd} + T_n) \cong O(n^2)$	4.1650 seconds	Population diversity	Lack of balance between phases
WOA	5	$O(T_{nd} + T_n) \cong O(n^2)$	3.2240 seconds	Fast convergence	Fall in a local optimum
SSA	2	$O(T_{nd} + T_n) \cong O(n^2)$	4.3560 seconds	Good convergence acceleration	Suffers from local Optima
ASO	2	$O(T_{nd} + T_{cn}) \cong O(n^2)$	4.9820 seconds	High Capability	Not good in global optima finding due to the diversity of distributions

7. Discussion

In solving global numerical optimization problems, the mathematical model of a system is adapted to the objective function, and the maximum or minimum values are tried to be found. As we face increasingly complex problems in the real world, developing efficient optimization techniques becomes more critical, essential and urgent than ever. Among the MH algorithms that meet this need most successfully are the methods in the SI category. Algorithms in this class are generally population-based and use swarm intelligence in the exploration and exploitation phases. They also have diversification and intensification features. In diversification, agents promise to find the global answer with their balanced behaviors between the phases with more frequent changes. In intensification, they try to converge quickly to the local answer. Therefore, it is necessary to keep the balance between these two features. Similarly, the balance between the exploration and exploitation phases is essential, and consequently, algorithms with these features are likely to exhibit successful performance in various problems. Accordingly, scientists and researchers are trying to present efficient optimization methods inspired by nature, mainly population-based algorithms.

The global exploration and local exploitation phases are critical for the search process in the ARO algorithm. The advantage of this nature-inspired algorithm is that each stage is simulated with a feature of rabbits. At the same time, it exhibits a balanced behavior between the two phases with its energy shrink feature. In the early stages of the search process, agents are deployed predominantly throughout the search area, and in later stages, they converge on the best candidate. The strengths of the ARO algorithm include stability, low cost, flexibility, easy implementation, simplicity, and no derivation mechanism. It uses stochastic operators, vital in escaping from local optima and converging to global optima. These features make it easily adaptable to different problems. This SI-based algorithm has many unique values, suggesting it will yield fruitful results when adapted to real-world problems. Therefore, the number of new studies on the ARO algorithm is increasing in the literature. These studies have been applied to multiple complex optimization problems (e.g., image processing, optimal scheduling, energy efficiency, system security, power systems, wireless communication systems, cancer diagnoses, stock market prediction, transportation systems, analysis of complex networks, etc.). On the other hand, the general shortcomings of the ARO algorithm can be listed as convergence accuracy, probability of falling into local optima, and lower population diversity. These problems have been mainly addressed in modified studies in literature. For this purpose, different studies have achieved

improvements using strategies such as chaotic maps, levy flight, selective opposition, Brownian random motion, and Gaussian mutation.

Comprehensive reviews of famous and successful MH algorithms in the literature, examining their strengths and weaknesses, and providing general information about the studies conducted are helpful for readers. The recent spread of such studies in the literature is promising and valuable [178-181]. Like other metaheuristic algorithms or modified versions [182-185], the ARO has advantages and some unavoidable shortcomings.

In this study, empirical simulations were performed on 20 test functions to analyze the specific features of ARO, and the results were compared with the well-known MH algorithms in the literature. According to the results, the ARO algorithm is among the algorithms that have shown exemplary performance in recent years. For these analyses, care was taken to ensure that the algorithms selected from the literature have different features. For example, while the SCSO is an algorithm that behaves balanced in the exploration and exploitation phases, the WOA algorithm is one of the excellent performance algorithms in exploitation. The MPA algorithm is a robust algorithm of recent years with a high-density ratio. The diversity of the selected algorithms allows the comparison to be more reliable and accurate. Moreover, in this paper, studies in the literature on ARO have been comprehensively examined in four different categories. These are improved, hybridization, variants of ARO, and adapted classifications. All these studies either used the algorithm's strengths to solve some complex issues or tried to overcome the algorithm's shortcomings. The investigated studies generally either used this algorithm as is, in a single or hybrid form, to solve various problems, or focused on some of the shortcomings of ARO and tried to improve them and solve different issues accordingly.

In the hybrid category, firstly, the combined operation of the ARO algorithm with different MH algorithms is discussed. The ABC, AO, GBO, GWO, WSO, HBA and TSO algorithms were used in the studies. More than 80% of these studies attempted to improve the exploration phase of ARO. The ARO algorithm was used as is in the survey conducted only with WSO. In this category, hybrid algorithms with ARO are also presented by including MH algorithms and ML methods. The GBO, HBA, SCSO, JSO and SSA algorithms were used as MH-based algorithms and NB, SVM, ANN, BP, CAT regression methods were used as ML approaches. There is a total of 5 studies that fit this class (14%). In the hybrid category, other subclasses include ML and DL based studies. Approximately 63% of the total studies belong to these two classes. Studies in the hybrid category constitute 31% of the total cited publications. General information about hybrid-based studies is presented in Tables 1 to 4.

We classified the other category with the variant name, and 8 of the 9 studies in this category are Multi-Objective Problems (MOP)-based studies. This indicates openness in different studies on binary ARO. In the MOP-based studies, researchers defined a suitable multi-objective structure for ARO due to the nature of the problems. Papers in this category constitute 9% of the total studies. Summary information is also presented in Table 5.

The studies collected in the improved category mainly focused on the slowness of the ARO algorithm in accuracy or speed of convergency, the lack of diversity in the population, and the possibility of getting stuck in local optima. These issues are mainly considered as disadvantages or shortcomings of ARO. In this regard, a total of 27 studies were conducted. They benefited from different MH algorithms or strategies such as chaotic, levy flight, Gaussian mutation, adaptive switching, and various learning techniques. These studies generally tried to control the drawbacks of random distributions or improved the functionality of the exploration/exploitation phases. In some of these studies, a single strategy was used, while in others, more than one method was used. Approximately 27% of the total studies were in this category. General information about the studies in this category is presented in Tables 6, 7, and 8.

The last category, Adapted, has the highest number of publications (33% of total citations). The standard (original) ARO algorithm is employed in the adapted class to solve a specific problem. In such studies, the strengths of the ARO algorithm were used as is. Studies in this category show the wide range of uses of the algorithm since they are in different branches of science. The ARO algorithm has mainly been used to solve problems such as finding optimal values for uncertain parameters or tuning hyperparameters. Table 9 provides an overview of the studies in this category.

Our study's analysis demonstrates that the ARO algorithm's original version can be adapted to many problems and indirectly provides the information that it has underlying robustness and good performance. In other words, research so far shows that the original ARO is as successful in finding solutions as the improved ARO, and therefore, the

basic structure of ARO is acceptable. In this way, it is considered an effective SI optimizer with a strong search ability and convergence rate for global optimization problems. On the other hand, modification studies to eliminate the ARO algorithm's shortcomings or increase its performance allow this algorithm to be more successful. Information that provides a general evaluation and discussion of the ARO algorithm, including advantages, shortcomings, and future predictions is presented in Table 13.

Table 13. Brief of advantages, shortcomings, and future insights of ARO algorithm.

Advantages	Shortcoming	Future insights
+ It has a balanced transition between the exploration and exploitation phases.	- Slow convergence	* Development of advanced versions or hybrids that can overcome the limitations of ARO and improve its performance.
+ It has simple structure and ease of adaptation.	- Probability of falling into local optima	* Benefits widespread of original or improved versions of the ARO algorithm in machine learning and deep learning-based studies.
+ It can adapt to a variety of problems because only a small number of parameters need to be adjusted.	- lower population diversity	* To optimize ARO's computational cost and time, designing parallel or distributed versions further.
+ Maintains stability in scenarios where there are disturbances.		* Using the current improved versions on a variety of untested complex problems.
+ Scalable for large optimization problems.		* Adaptation of this algorithm to problems in different interdisciplinary fields.
+ Quickly finds suitable solutions for time-sensitive problems.		* Providing further analysis studies on MH algorithms.
		* Further improving its strengths as possible.

8. Conclusion and Future Works

Artificial Rabbits Optimization (ARO), as a nature-inspired optimization algorithm, mainly has gained significant attention for their ability to tackle complex optimization problems by mimicking the behaviors of rabbits. This algorithm mathematically models the rabbits' survival strategies of circuitous foraging and random hiding. The foraging structure inspires the exploration phase of the algorithm, while the exploitation phase is based on the random hiding mechanism. In addition, it is envisaged to ensure balanced transitions between phases with the mechanism called energy shrink. This algorithm attracts great attention in the literature due to its advantages, such as its simple structure, ease of adaptation, and the use of few parameters. On the other hand, this algorithm suffers from weaknesses such as convergence accuracy, probability of falling into local optima, and lower population diversity. Many studies that take advantage of the strengths of this algorithm and address its shortcomings have been conducted in the last two years. Therefore, all ARO-based studies were examined in this study. This algorithm, which received a total of 248 citations, was used directly as a method in 101 studies. This paper concluded that most publications with citations to the ARO algorithm were published in Elsevier. Moreover, 73% of the total referencing studies were in the SCI index. In this comprehensive review, we evaluated the studies in four main categories. These are adapted, variants, hybrid, and improved classes. In the studies in the 'adapted' category, which allocated 33% of the citations, the standard ARO algorithm was used for various optimization problems. The problems in this field consist of multiple branches of science. The ARO algorithm has mainly been used to solve problems, such as finding optimal values for uncertain parameters or tuning hyperparameters. In this regard, suitable fitness functions for every problem have been defined. The class named 'variant' consists of binary and MOP subclasses, taking the least percentage share. A single binary-based study was conducted in this category. Studies in the other class, the 'hybrid' class, include studies based on other MH algorithms and/or ML-DL methods that interact with the ARO. Studies in this class constituted 31% of the total citations. The last class, 'improved,' consists of studies that try to improve or enhance various shortcomings of the ARO algorithm. The rate of studies in this class was 27%. Given the growing interest in ARO and its variants, this review aims to provide a comprehensive overview of the algorithm, its underlying principles, key components, and evolutionary trajectory. Moreover, the review thoroughly examines the various perspectives proposed in the literature, highlighting their strengths and limitations. In addition, the performance of the ARO algorithm was also analyzed numerically. In this regard, it was tested on 20 benchmark functions, and according to the results, the ARO algorithm was superior to other algorithms with a success rate of 85%. This review aims to provide valuable information for researchers and practitioners who want to utilize ARO in optimization problems in various fields, such as engineering, finance, health sciences, social sciences, etc., by

synthesizing the existing knowledge and identifying emerging trends.

Among the studies reviewed, studies focusing on the shortcomings of the algorithm are based on a narrow dimension; therefore, as future studies:

- Development of advanced versions or hybrids that can overcome the limitations of ARO and improve its performance.
- Benefits widespread of original or improved versions of the ARO algorithm in machine learning and deep learning-based studies.
- Further improving its strengths as much as possible.
- Designing parallel or distributed versions further to optimize ARO's computational cost and time.
- Using the current improved versions on a variety of untested complex problems.
- Adaptation of this algorithm to problems in different interdisciplinary fields.
- Providing further analysis studies on MH algorithms.

Appendix 1. Benchmark functions from CEC14 and CEC15.

Benchmark Function	Formula	dim	range	f_{min}	type
F1 Sphere	$f_1(x) = \sum_{i=1}^n x_i^2$	30	[-100,100]	0	Unimodal
F2 Schwefel 2.22	$f_2(x) = \sum_{i=1}^n x_i + \prod_{i=1}^n x_i $	30	[-10,10]	0	Unimodal
F3 Schwefel 1.2	$f_3(x) = \sum_{i=0}^{n-1} \left\{ \sum_{j=0}^{i-1} x_j \right\}^2$	30	[-100,100]	0	Unimodal
F4 Schwefel 2.21	$f_4(x) = \max_i \{ x_i , 1 \leq i \leq n \}$	30	[-100,100]	0	Unimodal
F5 Generalized Rosenbrock	$f_5(x) = \sum_{i=1}^{n-1} [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$	30	[-30,30]	0	Unimodal
F6 STEP	$f_6(x) = \sum_{i=1}^n ([x_i + 0.5])^2$	30	[-100,100]	0	Unimodal
F7 Quartic	$f_7(x) = \sum_{i=1}^n ix_i^4 + \text{random}[0,1)$	30	[-1.28,1.28]	0	Unimodal
F8 Generalized Schwefel	$f_8(x) = \sum_{i=1}^n -xi \sin(\sqrt{ x_i })$	30	[-500,500]	- 418.9829 *5	Multimodal
F9 Rastrigin	$f_9(x) = \sum_{i=1}^n [x_i^2 - 10 \cos(2\pi x_i) + 10]$	30	[-5.12, 5.12]	0	Multimodal
F10 Ackley	$f_{10}(x) = -20 \exp \left(-0.2 \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} \right) - \exp \left(\frac{1}{n} \sum_{i=1}^n \cos(2\pi x_i) \right) + 20 + e$	30	[-32,32]	0	Multimodal
F11 Griewank	$f_{11}(x) = \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1$	30	[-600,600]	0	Multimodal
F12 Generalized Penalized	$f_{12}(x) = \frac{\pi}{n} \left\{ 10 \sin(\pi y_1) + \sum_{i=1}^{n-1} (y_i - 1)^2 [1 + 10 \sin^2(\pi y_{i+1})] + (y_n - 1)^2 \right\} + \sum_{i=1}^n u(x_i, 10, 100, 4)$ $y_i = 1 + \frac{x_i + 1}{4}$ $u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m & x_i > a \\ 0 & -a < x_i < a \\ k(-x_i - a)^m & x_i < -a \end{cases}$	30	[-50,50]	0	Multimodal

F13	Generalized Penalized	$f_{13}(x) = 0.1 \left\{ \sin^2(3\pi x_1) + \sum_{i=1}^n (x_i - 1)^2 [1 + \sin^2(3\pi x_i + 1)] \right. \\ + (x_n - 1)^2 [1 + \sin^2(2\pi x_n)] \left. \right\} \\ + \sum_{i=1}^n u(x_i, 5, 100, 4)$	30	[-50,50]	0	Multimodal
F14	Shekel's Foxholes	$f_{14}(x) = \left(\frac{1}{500} + \sum_{j=1}^{25} \frac{1}{j + \sum_{i=1}^2 (x_i - a_{ij})^6} \right)^{-1}$	2	[-65,65]	1	Fixed dimension
F15	Kowalik's	$f_{15}(x) = \sum_{i=1}^{11} \left[a_i - \frac{x_1(b_i^2 + b_i x_2)}{b_i^2 + b_i x_3 + x_4} \right]^2$	4	[-5,5]	0.00030	Fixed dimension
F16	Six-Hump Camel-Back	$f_{16}(x) = 4x_1^2 - 2.1x_1^4 + \frac{1}{3}x_1^6 + x_1x_2 - 4x_2^2 + 4x_2^4$	2	[-5,5]	-1.0316	Fixed dimension
F17	Branin	$f_{17}(x) = \left(x_2 - \frac{5.1}{4\pi^2} x_1^2 + \frac{5}{\pi} x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos x_1 + 10$	2	[-5,5]	0.398	Fixed dimension
F18	Goldstein Price	$f_{18}(x) = [1 + (x_1 + x_2 + 1)^2 \\ \times (19 - 14x_1 + 3x_1^2 - 14x_2 + 6x_1x_2 \\ + 3x_2^2)]. [30 \\ + (2x_1 - 3x_2)^2 \\ \times (18 - 32x_1 + 12x_1^2 + 48x_2 - 36x_1x_2 \\ + 27x_2^2)]$	2	[-2,2]	3	Fixed dimension
F19	Hartman's Family	$f_{19}(x) = - \sum_{i=1}^4 c_i \exp \left(- \sum_{j=1}^3 a_{ij} (x_j - p_{ij})^2 \right)$	3	[1, 3]	-3.86	Fixed dimension
F20	Hartman's Family	$f_{20}(x) = - \sum_{i=1}^4 c_i \exp \left(- \sum_{j=1}^6 a_{ij} (x_j - p_{ij})^2 \right)$	6	[0, 1]	-3.32	Fixed dimension

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